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Natural Resource Consultants

**2003 Lower Mayo River
Chinook and Channel Assessment**

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Prepared for:

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ABSTRACT

Mayo River is an important chinook spawning and rearing stream. The construction of Wareham Dam in 1952 has prevented adult and juvenile fish passage into the spawning and rearing habitats upstream of the dam as well as regulating the flows downstream of the dam. An analysis of historic air photos for the area below Wareham Dam suggests that construction and possibly the operation of the dam has reduced the length of secondary channels. Similar changes have been documented on numerous other regulated rivers. The reduction of secondary channels, is thought to decrease the rearing opportunities for juvenile chinook salmon. Opportunities for mitigating these impacts through the restoration of side and back channel habitats or developing groundwater fed channels were investigated. Minnow trapping was conducted to determine rearing dynamics of juvenile chinook in Mayo River to help guide the types of habitats that should be restored. This report describes a number of identified restoration sites, including two that have a high likelihood of success as they have a reliable water source and they are presently used by juvenile chinook. The implementation of this type of habitat restoration has not been undertaken in Yukon. The results of the proposed works will provide a basis for assessing the effectiveness of this type of habitat restoration work.

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Several staff members of the First Nation of Nacho Nyak Dun, Mayo, contributed to this project. Millie Olsen, GIS Technician, Lands and Resources Department, completed some of the maps present in this report. Dick Mahoney and Ann Leckie assisted throughout the project. Roman Krska, Dick Mahoney and Crystal Stevenson helped with the various field components.

Nathan Ferguson from Environmental Dynamics Inc. assisted in the field components of the project, conducted the minnow trapping survey and prepared some of the maps in this report. William Patterson, a local resident of Mayo, assisted with the minnow-trapping program.

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Ms. Elizabeth Goldsworthy, B.Sc., and Ms. Sandy Allegretto, B.Sc., of M. Miles and Associates Ltd. undertook the hydrologic analyses and prepared the historical air photo compilations.

STATEMENT OF LIMITATIONS

This document has been prepared by EDI Environmental Dynamics Inc. and M. Miles and Associates Ltd. for the exclusive use and benefit of the Yukon River Panel and First Nation of the Nacho Nyak Dun. No other party is entitled to rely on any of the conclusions, data, opinions, or any other information contained in this document.

This document represents EDI's and MMA's best professional judgement based on the information available at the time of its completion and as appropriate for the project scope of work. Services performed in developing the content of this document have been conducted in a manner consistent with that level and skill ordinarily exercised by biologists and geoscientists currently practising under similar conditions. No warranty, expressed or implied, is made.

TABLE OF CONTENTS

	Page
ABSTRACT	i
ACKNOWLEDGEMENTS	ii
STATEMENT OF LIMITATIONS	ii
TABLE OF CONTENTS	iii
LIST OF TABLES	iv
LIST OF PHOTOS	iv
1.0 INTRODUCTION	1
1.1 Project Scope	1
1.2 Project Objectives	1
2.0 LOCATION OF STUDY AREA	1
3.0 METHODOLOGY	2
3.1 Review of Existing Information.....	2
3.2 Air Photo Analysis	2
3.3 Field Investigations	2
4.0 ENVIRONMENTAL SETTING	2
4.1 Climatology.....	2
4.2 Hydrology	3
4.2.1 Available Information	3
4.2.2 Seasonal Variation in Discharge	5
4.2.3 Historical Variation in Peak Flow.....	5
4.2.4 Comparison of Pre- and Post-Regulation Flows	5
4.3 Channel Processes.....	6
4.3.1 Reach Description.....	6
4.3.2 Channel Stability	7
4.4 Fisheries Resources	8
5.0 FIELD INVESTIGATIONS	9
5.1 Reach 1.....	9
5.2 Reach 5.....	12
6.0 RESTORATION OPPORTUNITIES	13
6.1 Objectives.....	13
6.2 Possible Restoration Sites.....	14
6.2.1 Site A - Potential Groundwater Channel.....	14
6.2.2 Site B. – Side Channel with Seasonal Flow.....	16
6.2.3 Site C. – Existing High Water Back Channel.....	17
6.2.4 Site D – Former Gravel Pit.....	19
6.2.5 Site E – Existing Groundwater Channel.....	20
7.0 CONCLUSIONS	22
8.0 RECOMMENDATIONS	23
8.1 Physical Studies.....	23
8.2 Biological Studies	23
9.0 LITERATURE CITED	24
APPENDIX A: Figures	27
APPENDIX B: Minnow Trap Data	28

LIST OF TABLES

Table 4.2.1	Summary of Available Discharge Data.....	4
Table 4.2.2	Average return period of unusually large floods since 1959.....	5
Table 4.4.1	Fish species documented in the Mayo River watershed.	8

LIST OF PHOTOS

	Page
Photo 1.	Upstream view of Reach 1 showing typical glide habitat. 10
Photo 2.	Downstream view of pool within a side channel with a significant amount of cover..... 11
Photo 3.	Typical Back channel Habitat..... 12
Photo 4.	Downstream view of area immediately downstream of road crossing..... 15
Photo 5.	Upstream view of a small seasonal drainage near the confluence with the Mayo River. 15
Photo 6.	Upstream view of side channel which had very little water at time of survey (July 27, 2003)..... 16
Photo 7.	Downstream view of isolated pool (July 27, 2003). 17
Photo 8.	Downstream view of high-water back channel area..... 18
Photo 9.	Upstream view of back channel taken from confluence with Mayo mainstem. . 18
Photo 10.	Overview looking downstream in July..... 19
Photo 11.	Cross view of pond in September. 20
Photo 12.	Upstream view of existing groundwater channel. 21
Photo 13.	Downstream view of groundwater channel. 21

LIST OF FIGURES PRESENT IN APPENDIX A

Figure 2.1	Location of study area within the Mayo River Watershed.
Figure 4.1.1	Map showing locations of hydrometric and climate stations.
Figure 4.1.2	Seasonal variation in temperature at Mayo Airport.
Figure 4.1.3	Seasonal variation in mean monthly precipitation at Mayo Airport.
Figure 4.1.4	Seasonal variation in mean snow depth at Mayo Airport.
Figure 4.2.1	Seasonal variation in discharge or water level.
Figure 4.2.2	Historical variation in annual maximum daily discharge or water level.
Figure 4.2.3	Annual flow duration curve for natural and regulated flow regimes.
Figure 4.3.2	1996 air photo mosaic of reaches 5,4,3 and 2.
Figure 4.3.3	1996 air photo mosaic of reaches 2 and 1.
Figure 4.3.4	1996 air photo mosaic of Reach 1.
Figure 4.3.5	Historical changes in channel morphology (reaches 5, 4 and 3), Mayo River.
Figure 4.3.6	Historical changes in channel morphology (reaches 3 and 2), Mayo River.
Figure 4.3.7	Historical changes in channel morphology (reaches 2 and 1), Mayo River.
Figure 4.3.8	Historical changes in channel morphology (Reach 1), Mayo River.
Figure 4.4.1	Average catch rate of juvenile chinook per hour per minnow trap in 1994.
Figure 5.1.1	Minnow trap locations and results for the Mayo River for late July 2003.
Figure 5.1.2	Minnow trap locations and results on the Mayo River for late September 2003.
Figure 5.1.3	Catch rate within each habitat type.

1.0 INTRODUCTION

1.1 Project Scope

Mayo River¹ is a significant chinook salmon (*Oncorhynchus tshawytscha*) spawning and rearing stream in the Stewart River Drainage. However, construction of the Wareham Dam in the early 1950's has significantly reduced the habitat accessible to salmon. This dam, which is located approximately 12 km upstream of the Stewart River confluence, prevents upstream migration and this has reduced the amount of spawning and rearing habitat available to chinook salmon. Buchan (1993) compiled local and traditional knowledge which indicated that chinook salmon were historically observed as far upstream as Roop Lake located upstream of Mayo Lake. This study also found a report of hundreds of king (chinook) salmon swimming around below the dam the year following construction. Kendel (1973) noted that, prior to the construction of Wareham Dam, the main spawning grounds were located at the outlet of Mayo Lake.

As the capacity of Mayo River for salmon has decreased significantly since the construction of the dam, ensuring that remaining habitat is in an optimum state is essential to maintaining the Mayo River chinook salmon population. This project evaluates the habitat downstream of the dam and examines restoration or improvement opportunities for chinook salmon.

1.2 Project Objectives

The objectives of this project include:

1. Assessing the physical habitat changes that have occurred on lower Mayo River following construction of Mayo and Wareham Dams;
2. Assessing chinook habitat and habitat utilization within the lower Mayo River and tributaries;
3. Identifying opportunities to improve habitat for adult and juvenile chinook salmon; and
4. Providing training and employment, building technical capacity and fostering stewardship for Nacho Nyak Dun [NND] citizens.

2.0 LOCATION OF STUDY AREA

The Mayo River drains a watershed of approximately 2,640 km² and flows into Stewart River near the community of Mayo (Figure 2.1). Mayo River is currently dammed in two locations, directly downstream of Mayo Lake and approximately 12 km upstream from the mouth (Wareham Dam). The project study area consists of the 12 km section of channel downstream of Wareham Dam which is currently accessible to salmon.

¹ (Watershed code: 832-00000)

3.0 METHODOLOGY

The study was conducted in a number of steps, which are summarized below.

3.1 Review of Existing Information

Biologists from EDI Environmental Dynamics Inc. (*Environmental Dynamics*) and geomorphologists from M. Miles and Associates Ltd. (*MMA*) compiled existing literature and data that had been conducted by government agencies, consultants and developers. Information pertaining to chinook salmon and spawning documentation was recorded and mapped where applicable. Flow data collected by Water Survey of Canada and Yukon Energy Corporation was also evaluated.

3.2 Air Photo Analysis

An air photograph mosaic was prepared for the study area using the most current (1996) air photographs. Additional photographs taken in 1949, 1966 and 1984 were also compiled to document changes in channel conditions over time. Air photograph interpretation was also conducted to identify potential groundwater fed restoration sites and prioritize areas which needed to be assessed in the field.

3.3 Field Investigations

Biologists, a geomorphologist and NND staff conducted primary field investigations between July 27 and 28, 2003. This work assessed river conditions and associated fish habitat. Potential restoration sites were also inspected and their ability to provide improved juvenile chinook habitat was evaluated.

More detailed fieldwork, including minnow trapping to evaluate the habitats being used by juvenile chinook, was undertaken in late July and late September. Minnow trapping was completed in a variety of habitat types following the protocol developed by Department of Fisheries and Oceans (1994).

Field studies were primarily undertaken in Reaches 1 and 5 as access difficulties in other portions of the study area effectively preclude mechanized habitat restoration activities.

4.0 ENVIRONMENTAL SETTING

4.1 Climatology

The Meteorological Service of Canada (MSC) operates a climate station at Mayo Airport. This site is located on the east side of the Mayo River Valley approximately 3 km from the confluence with Stewart River (Figure 4.1.1). The elevation of Mayo Airport is 504 m which is similar to the elevation of the adjacent section of Mayo River. Given this similarity and close proximity, the data from Mayo Airport provide a good indication of the climatic conditions in the study area.

The seasonal variation in temperature, based on 1971 to 2000 normals, is shown on Figure 4.1.2. This analysis indicates that the average annual air temperature is -3.1°C . Average monthly air temperatures exceed 0°C in the six month period between April and September. The extreme minimum temperature of -62.2°C occurred in February and the maximum temperature of $+36.1^{\circ}\text{C}$ occurred in June. The

seasonal variation in precipitation, based on 1971 to 2000 normals, is illustrated on Figure 4.1.3. The average annual precipitation is 205 mm of which 47% occurs as snow. The maximum average monthly precipitation of 54.4 mm occurs in July, while April has the minimum average total of 9.2 mm.

The average seasonal variation in snow cover at Mayo Airport is illustrated on Figure 4.1.4. The snow free period typically extends between May and September. The average maximum snow depth of 38 cm occurs in February.

Additional information on the climatology of the study area, and the effect of changing climatic conditions, is available in reports by Wahl *et al.* (1987); McCoy and Burn (2001) and Anonymous (2001).

4.2 Hydrology

4.2.1 Available Information

Stream discharge data have been collected at four sites in the Mayo River watershed (*Table 4.2.1*). The Water Survey of Canada (WSC) operated the station *Mayo River Near Mayo* between 1945 and 1950. Data from this site, which was located upstream from Wareham Lake, predates construction of Wareham Dam. The gauged basin area was 2,260 km².

The WSC, in cooperation with the Yukon Energy Corporation, has been collecting discharge data from Wareham Dam since 1958. This station, which is called *Wareham Lake at Headgate*, records both the amount of water spilled over the structure and flows passed through the turbine. These data have been recently re-compiled for the Yukon Energy Commission and a preliminary version has been made available by Mr. Cord Hamilton, P.Eng. (pers. comm.).

There are two other sources of hydrometric data. The WSC has monitored water levels on *Mayo Lake Near The Outlet* since 1980 and Indian and Northern Affairs Canada (INAC) operated a seasonal gauging station on *Duncan Creek at Mayo Lake Road* between 1979 and 1982. Duncan Creek is a headwater tributary to Mayo River. The basin area at the gauging station is 228 km².

The WSC has also operated two gauging stations on Stewart River. *Stewart River at Mayo* operated from 1947 to 1979. *Stewart River near Mayo* was established at the same location in 1980, however, the station now only reports water levels. Nevertheless, these data are useful when evaluating the effects of Stewart River flows on the lowermost section of Mayo River. The locations of the above stations are indicated on Figure 4.1.1.

TABLE 4.2.1: SUMMARY OF AVAILABLE DISCHARGE DATA

STATION NUMBER	STATION NAME	BASIN AREA (km2)	TYPE OF RECORD	TYPE OF FLOW	YEARS OF RECORD ANNUAL MAXIMUM DAILY DISCHARGE OR LEVEL
09DC001	Mayo River near Mayo	2,260	1945/47 MS; 1948 #; 1949/50 MS; 1951/53 #	REG	4
09DC002	Stewart River at Mayo	31,600	1947/48 M#; 1949/51 MC; 1952/55 MS; 1956/56 MC; 1957-58 MS; 1959/71 MC; 1972/79 RC	NAT	31
09DC004	Wareham Lake at Headgate	--	1979/2002 *RC	REG 52	44
09DC005	Mayo Lake near the Outlet	--	1979/2002 *RC [water level]	REG 52	21
09DC006	Stewart River Near Mayo ^{*A}	--	1980-2002 *RS [water level]	NAT	22
29DC001	Duncan Creek at Mayo Lake Road	228	1979-1982 R	Part REG	4

Inland Waters Directorate, 1996

NOTES:

* stage only
 # miscellaneous measurements
 M manual gauge
 R recording gauge
 S seasonal operation
 C continuous operation
 numbers refer to years (e.g. 09 is 1909)

REV Data to 19__ have been reviewed
 NAT Natural Flow
 REG Regulated Flow with date of regulation if known
 NA Not Available

(i) Data not published
 1 Daily Flows
 2 Instantaneous Flows
 7 Satellite D.C.P.
 10 Minimum Flows

***A** station at same location as DC002, however, levels only from 1980

4.2.2 Seasonal Variation in Discharge

The seasonal variation in discharge observed at each of the four gauging stations in the Mayo watershed is illustrated on Figure 4.2.1. This analysis indicates that the spring snowmelt freshet, which begins in mid-April, is the dominant hydrological event of the year on Mayo River. Summer and early-fall rainstorms can result in comparatively modest increases in flow until late-September when stream flow begins to rapidly drop to mid-winter base flow values.

There are three major tributaries to Mayo River downstream of Mayo Lake. Flow on Minto Creek is naturally regulated or 'buffered' by Minto Lake. In contrast, there are no large lakes on Duncan and Davidson Creek. The limited discharge data from Duncan Creek indicate that mid-summer rainstorm events can be larger than the flows which occur in the latter part of the freshet period. Summer rainstorms could therefore have the potential to result in sizeable flows on 'unbuffered' tributary streams.

4.2.3 Historical Variation in Peak Flow

The historical variation in annual maximum daily peak flows is illustrated on Figure 4.2.2. A flood frequency analysis has been undertaken on the data from *Wareham Lake at Headgate*.¹ This compilation indicates that four unusually large floods have occurred in the period since 1959 (Table 4.2.2).

Table 4.2.2 Average return period of unusually large floods since 1959.

YEAR	AVERAGE RETURN PERIOD (years)
1964	20
1972	25
1975	35
1991	50

The 1975 and 1991 floods are sufficiently large that they could have had a noticeable effect on channel stability.

4.2.4 Comparison of Pre- and Post-Regulation Flows

Comparison of the four years of pre-regulation discharge data from *Mayo River Near Mayo* with the regulated data from *Wareham Lake at Headgate*, suggests that flow regulation has reduced the size of the annual flood peaks. The average annual maximum pre- and post-regulation daily discharges are 93 and 74 m³/s, respectively. A comparison of flow duration data is presented on Figure 4.2.3. This analysis suggests that river regulation has reduced the frequency of flows of greater than 20 m³/s and increased the frequency of smaller discharges. This is a common effect on rivers with a storage reservoir

1 A flood frequently analysis of regulated data is subject to considerable uncertainty. These calculations are therefore preliminary and must not be used as a basis for design.

Unfortunately the period of pre-regulation data is amply short that the reliability of the above comparisons is unknown. The best way of overcoming the lack of pre-project data would be to undertake a detailed analysis of the operating characteristics of both Wareham and Mayo Lake Dams. Specifically, discharge and lake level data could be used to re-construct natural inflow volumes. Some of this data is presently available, however, a thorough analysis is beyond the scope of the present assignment.

4.3 Channel Processes

4.3.1 Reach Description

Previous work by Triton (1992) identified five river reaches in the area downstream of Wareham Dam and this delineation has been adopted for the purposes of consistency¹ (Figure 4.3.1). Air photo mosaics, showing channel conditions in 1996, are presented on figures 4.3.2 to 4.3.4.

Reach 5 consists of a narrow confined channel in the area between Wareham Dam and a point just downstream of the powerhouse. This 0.7 km long reach has a cobble and gravel bed.

Reach 4 extends 2.4 km downstream of the powerhouse. This irregular channel is predominantly single thread. A number of point bars and small islands occur in areas where the channel is not laterally confined. There is a large area of slope instability on the right bank at Km 1.7.²

Reach 3 consists of a sinuous, single thread, laterally confined channel. This reach is 1.5 km long.

Reach 2 is the first alluvial section downstream of Wareham Dam. This irregularly meandering, 2.5 km long, gravel bed channel contains a number of vegetated islands and a variety of secondary channels or wetlands. There is a wide valley flat which is bounded by a series of fluvial terraces. The location of the break between Reaches 1 and 2 appears to have been arbitrarily chosen as there is not an abrupt change in channel characteristics.

Reach 1 consists of a 5 km long, irregularly meandering, gravel bed channel flowing within a wide valley flat. Numerous point bars, islands, secondary channels and wetlands have been formed as a result of channel shifting. A highway bridge crosses the channel at Km 8.8 and a buried sewer line is located at Km 11.3. An elevated setback dyke has also been constructed along the lower left bank³ to reduce flooding hazard to the town of Mayo.

1 The numbering of the reaches have been changed to reflect the conventional method of numbering reaches from the mouth of a stream in a upstream direction.

2 Stream kilometres have been measured downstream of Wareham Dam.

3 Looking in a downstream direction.

4.3.2 Channel Stability

Historical air photos taken in 1949, 1966, 1984 and 1996 have been compiled on Figures 4.3.5 to 4.3.8 to illustrate changes in channel stability over time. This analysis indicates that the construction of Wareham Dam has had a significant affect on the morphology and stability of lower Mayo River. Much of this impact has resulted from increased sediment loadings which occurred during construction of the earth fill dam and powerhouse. Flow regulation (specifically a reduction in flood peaks) commonly reduces the number of side and back channels (e.g. Simons, et al., 1981; Kellerhals, 1982; Williams and Wolman, 1984; Kellerhals and Church, 1989; Church, 1995 and Brandt, 2000). A similar effect may also be occurring on lower Mayo River. However, this would need to be confirmed by more detailed hydrological and field investigations.

A photograph in a pamphlet produced by Yukon Energy (no date) suggests that dam construction and site preparation for the powerhouse resulted in the introduction of extensive quantities of sediment into Reach 5. This observation is supported by the 1966 air photos which show extensive areas of slope disturbance in the vicinity of the dam and powerhouse. However, post-1949 changes in channel morphology in Reach 5 are limited due to the steep confined nature of the channel.

The number and extent of instream gravel accumulations dramatically increased in Reach 4 in the period between 1949 and 1966. This is thought to reflect sediment production from construction of Wareham Dam and the powerhouse in 1951/1952. Most of these gravel deposits have subsequently revegetated. A number of previously active secondary channels (such as the side channel at Km 2.0) have also been abandoned over the period between 1949 and 1996. This could be the result of both construction related sediment accumulation and reduced peak flows due to dams on Wareham and Mayo Lakes.

Reach 3 has showed little change in morphology over the period since 1949. This is expected as the confined single thread channel has only a limited ability to store sediment.

The alluvial Reach 2 was heavily infilled with sediment in the period between 1949 and 1966. This resulted in extensive channel shifting and many infilled side channels (e.g. the left bank at Km 7.4) have been subsequently abandoned. Most of these sediment deposits are now vegetated and the channel has developed a more regular meander configuration over the period since 1966. Large flood flows in 1975 and 1991 likely expedited this recovery process.

The alluvial Reach 1 has undergone a similar pattern of post-1949 sediment deposition followed by slow channel recovery. The amount of sediment deposition was, however, more widespread than in Reach 2. [This may reflect a lower channel gradient.] A sizeable, likely man made, channel cut off also occurs immediately upstream of the highway bridge. This has resulted in channel instability in the adjacent area. The pipeline crossing at Km 11.4 has partially infilled a series of secondary channels on the right bank flood plain and this may be increasing the rate of sediment deposition and vegetation encroachment in these small channels. In contrast, gravel excavation (possibly associated with the construction of the left bank set back dyke) has resulted in the formation of a sizeable wetland area near the town of Mayo. Post-construction channel progression has recently connected this area to the mainstem river.

These preliminary analyses indicate that anthropogenic activities have resulted in extensive changes to the morphology and stability of lower Mayo River. Sediment deposition and, possibly, changes in the

flow regime appear to have had the greatest impact. However, significant local effects have also occurred as a result of bridge, pipeline and dyke construction. Our analysis indicates that a number of secondary channels have been abandoned since 1949. It would be interesting to locate older maps or other information showing conditions earlier in the century. It is possible that mining-related valley flat and upslope logging activities caused extensive channel instability in the late 19th or early 20th century. The present trend towards a loss of side and back channels might, therefore, partially reflect recovery processes arising from these earlier impacts.

4.4 Fisheries Resources

Twelve species of fish have been documented within the Mayo River watershed (Table 4.4.1).

Table 4.4.1 Fish species documented in the Mayo River watershed (as documented by FISS, 2003).

Fish Species Common Name	Scientific Name
chinook salmon	<i>Oncorhynchus tshawytscha</i>
chum salmon ¹	<i>Oncorhynchus keta</i>
Arctic grayling	<i>Thymallus arcticus</i>
round whitefish	<i>Prosopium cylindraceum</i>
lake whitefish	<i>Coregonus clupeaformis</i>
burbot	<i>Lota lota</i>
Inconnu	<i>Stenodus leucichthys</i>
northern pike	<i>Esox lucius</i>
lake trout	<i>Salvelinus namaycush</i>
slimy sculpin	<i>Cottus cognatus</i>
longnose sucker	<i>Catostomus catostomus</i>
least cisco	<i>Coregonus sardinella</i>

¹ As listed by Buchan 1993; please note FISS 2003 also documents the presence of chum salmon; however, after further investigation no record of chum was found for the reference listed in FISS.

The entire section of the Mayo River between Wareham Dam and the mouth appears to be very productive in terms of chinook habitat. There are two main documented spawning areas, one in Reach 2 and one in Reach 4 (Triton 1992; Figure 4.3.1). Estimates and counts of spawning chinook indicate that the Mayo River is a significant spawning population in the Stewart River Watershed. Triton (1992) made a rough population estimate (based on the number of carcasses recovered) of an escapement of between 588-1940. In 1994, a weir was constructed and operated on the Mayo River and 642 adult chinook passed through between July 31 and August 28th (Heron 1994).

Minnow trapping for juvenile chinook was also conducted throughout the lower portions of the Mayo River in 1994. Sampling results indicate that growth rates for juvenile chinook averaged 0.5 mm / day and 0.04 g / day between June 15 and July 20, 1994 (derived from Heron 1994). Using this 1994 minnow trapping information the average catch rate of juvenile chinook was determined during different dates throughout the summer. The results show higher catch rates towards the later part of the year (Figure 4.4.1). This is thought to result from lower water levels. It should be noted that the majority of habitats sampled during this study appeared to be either mainstem or very large side channels.

Triton (1992) indicated that the area between the spillway and the powerhouse (Reach 5) is generally dry during the winter months with the exception of groundwater infiltration or seepage. Triton also

documented the apparent presence of redds below the dam in 1992 following a reduction of flows. During the latest water re-licensing process in 2000, the Yukon Salmon Committee expressed concern about the possible dewatering of Reach 5 on Mayo River. To address this concern, Yukon Energy indicated that they were prepared to investigate salmon utilization in this section of stream (McWilliam 2000). The water license (HY99-012) that was renewed and is effective from January 1, 2001 to December 31, 2025 does not include minimum requirements for releasing water over the spillway. A review of recent flow data indicates that from January 2001 to May of 2003 indicated that there was flow over the spillway of at least 48 cubic feet per second or 1.36 m³/sec.

A transmission line between Mayo and Dawson was completed in late 2003. This new system has increased the power demand on Wareham Dam and changed the operation of the hydro system (Hamilton pers. comm. 2003). Specifically, additional water will be stored over the summer and flows will be increased during the winter. However, the exact nature of the changes cannot be fully predicted at this time.

5.0 FIELD INVESTIGATIONS

5.1 Reach 1

The lower Reach (Reach 1) of the Mayo River, is a low gradient (0.2%; according to Triton 1992) unconfined stream and contains a variety of habitats. These include mainstem, side channel and back channel habitat. Slaney and Zaldokas (1997) define a side channel as a lateral channel (smaller than the mainstem) with an axis of flow roughly parallel to the mainstem, which is fed by water from the mainstem. For the purposes of this report a back channel is defined as an area within the floodplain that is connected and accessible (at least seasonally) to the main or side channels via back flooding.

The mainstem includes riffles, glides and pools with a variety of cover types occurring along the margins of the stream (Photo 1).



Photo 1. Upstream view of Reach 1 showing typical glide habitat.

Side channel habitats range from small to large (5 to 40 m wide). The smaller side channels (<15 m channel width) provide habitats similar to small tributary streams in that they have significant amounts of cover due to the influence of the surrounding forest (Photo 2). The amount of usable habitat in many of these side channels was directly influenced by water levels. Many side channels were dry or only had isolated pockets of water during low flows in July; however, they provided much more suitable rearing habitat during increased flows in September.



Photo 2. Downstream view of pool within a side channel with a significant amount of cover (woody debris; September 29).

Back channel habitats are scattered throughout the flood plain in Reach 1 (Photo 3). These channels range in elevation and the amount of usable back channel habitat is dependant on water levels. During high flows most back channels will be accessible by fish.



Photo 3. Typical Back channel Habitat.

Results of minnow trapping in late July and late September are presented on figures 5.1.1 and 5.1.2 respectively. The results of sampling as per habitat type (main, side and back channel) are summarized in Figure 5.1.3. In general, catch rates were highest in mainstem and side channel habitat in the summer. However, in the fall juvenile chinook utilized all habitat types including back channels. While Figure 5.1.3 indicates mainstem habitats had the highest catch rate in late July, closer investigation of the data indicates that side channels with a significant amount of flow had a catch rate of 0.73 chinook per hour which was equal to that of the mainstem channel.

While no sampling was performed in the spring, it can be inferred that juvenile chinook were using some back channel habitat as several juvenile chinook were observed trapped in a back channel that had become isolated due to low water levels (near Trap 6; Figure 5.1.1). Based on size these were underyearling (0+) chinook that likely moved in to these habitats during high flows in the spring.

5.2 Reach 5

Reach 5 was dominated by very fast and turbulent flow, cobbles and boulder dominated substrates with few slower moving areas that would be suitable for adult holding or spawning. Field assessments below the dam found no chinook spawners on July 28 and approximately 5 to 7 on August 13. One pair was observed in the pool directly downstream of the spillway and 3 to 5 adults were observed in a slower moving run section (Mahoney pers. comm. 2003).

On December 16, 2003, Nacho Nyak Dun, Lands and Resources staff visited Reach 5 to get a feel for the water levels and flows. Observations were that flows over the spillway were about <5 % of the

flows downstream of the powerhouse; however, there appeared to be evidence of flow in most places throughout the reach (Mahoney pers. comm. 2003).

6.0 RESTORATION OPPORTUNITIES

6.1 Objectives

Studies within the Yukon River drainage indicate that underyearling (0+) chinook often move up into small non-natal streams for rearing purposes (Bradford *et al.* 2001). Presumably before the construction of the Wareham Dam, chinook spawned in the Mayo River could migrate into many non-natal tributaries upstream of the dam. With the possible exception of one, small second order (1:50,000) tributary stream to the lower Mayo River, juvenile chinook that do not migrate out of the Mayo system are likely limited to habitats along the mainstem Mayo River. The juvenile rearing conditions within this area are therefore thought to be critical to the success of the present Mayo River chinook salmon run.

Mayo River is extensively used by juvenile chinook salmon for rearing purposes; however, the amount of habitat available, and the use of habitats, appears to be dependant on flow levels and time of year. When flows are low (in the summer and winter), the amount of habitat available appears to be reduced to the mainstem and larger side channels. The 2003 minnow trapping program found that, during summer low flows, the majority of chinook were captured in mainstem and major side channel areas with significant flow. The 1994 trapping data of main channel habitats appears to indicate that more fish (as indicated by higher catch rates) are moving into these habitats as flows decrease. Back channel and smaller side channels habitats appear to be used in the spring and fall as refuge habitat from higher flows.

Predatory species such as pike (observed and captured in late July; Appendix B), high water temperatures during the summer (measured as high as 19 °C during July minnow trapping) and the limited amount of cover appear to limit the quality of main channel habitat during the summer low flow period. In general, riparian vegetation has a greater influence on morphology, cover and shading on smaller streams or channels compared to larger streams such as the Mayo mainstem. For example, in terms of morphology, Mossop (2003) found while studying streams in the Yukon with channel widths of 3.4 to 7.7 m that large woody debris was important in forming 28% of the pools and that chinook densities were correlated with large woody debris abundance. Also, as pike prefer large slow moving water bodies, they are generally not found in the flowing waters of smaller streams.

The apparent post-dam reduction in the number and length of side and back channels documented during the air photo analysis indicates that the rearing potential in the lower portion of the Mayo River appears to have declined following regulation. Restoration works and strategies to offset this apparent loss of habitat appear necessary.

It might be possible to flush out some of the infilling secondary channels (or create new ones) by releasing a sizeable flood flow from both the Mayo and Wareham Lake Reservoirs. As discussed in Webb *et al* (2000), controlled floods have the potential to mobilize sediments and, at least locally, re-establish pre-regulation channel morphologies. More research would, however, be needed to determine the magnitude and duration of the required flows, plus the ability of the two reservoirs to generate this volume of discharge.

An alternative, although likely short term, method of increasing the abundance of secondary channels would be to locally lower sections of existing channel. The objective would be to increase sub-surface inflow during the low flow period. The influx of river sediment would be minimized by limiting the work to the downstream portion of the channel [i.e. the intent would be to avoid work in the upstream inlet (should one exist), such that the potential for channel downcutting or enlargement is minimized].

The development and rehabilitation of off-channel areas is receiving increased attention as a practical means of restoring salmonid fish habitat, especially where main channel restoration is impractical (Lister and Finnigan 1997). In general, this type of restoration has successfully provided spawning and rearing habitat for both coho salmon (*O. kisutch*) and chum salmon (*O. keta*). While chinook do not spawn in off-channel habitat, some interior British Columbia stocks make use of off-channel ponds and side channels associated with tributaries for juvenile rearing and overwintering (Anonymous 1987 and Swales and Levings 1989 in Lister and Finnigan 1997).

In terms of restoration projects conducted for juvenile chinook, Richards and Cernera (1992) found that creating off channel habitat by linking up isolated dredge ponds (from past mining operations) on a river in Idaho substantially augmented the habitat available for juvenile chinook salmon. This study found that while hatchery stocked juvenile chinook used both channels and ponds, there was a strong preference for channel habitat with cover and flow. The minnow trapping results on the existing habitats for the Mayo River for juveniles would indicate a similar pattern, especially during the low flow sampling period in late July.

6.2 Possible Restoration Sites

A number of the inspected sites had the potential for restoration works. These include back channel, groundwater and side channel habitats. A description of the sites, possible methods of restoring or improving these sites and the merits of each are described in the following sections. Site locations are indicated on figures 6.2.1 and 6.2.2.

6.2.1 Site A - Potential Groundwater Channel

This drainage originates along the base of a hill upstream of the road to the sewage facility (Figure 6.2.1). This old Mayo River side channel now consists of a depression with no evidence of significant flow (Photos 4 and 5). Deepening the section of wall-based channel above the local road could possibly intercept sub-surface flow. The downstream portion of the channel is presently periodically wetted during high water. Local excavation within this channel could also increase subsurface inflows and provide additional water depths during the low flow period. There is an 80 cm diameter culvert at the road crossing which would have to be replaced if a groundwater channel was constructed in this area. The main disadvantage of this site is that the amount and level of groundwater is not known. Construction of a channel would not guarantee that enough groundwater would be recruited to provide fish habitat.



Photo 4. Downstream view of area immediately downstream of road crossing.



Photo 5. Upstream view of a small seasonal drainage near the confluence with the Mayo River.

6.2.2 Site B. – Side Channel with Seasonal Flow

This side channel is approximately 300 m long and flows through a forested section of flood plain (Figure 6.2.1). The unvegetated channel is approximately 10 m wide. This area was mostly dry during the July survey, with the exception of some deep isolated pools (Photos 6 and 7). Juvenile chinook were observed in pools and small pockets of water along the side of the channel. These fish would remain trapped until higher flows connected the pools and pockets of water to the main channel. The water temperatures within the isolated deep pool habitats were cool (5.5 °C on July 27th) indicating that groundwater inflow was occurring.



Photo 6. Upstream view of side channel which had very little water at time of survey (July 27, 2003).



Photo 7. Downstream view of isolated pool (July 27, 2003).

Lowering a portion of this side channel by approximately 1 to 2 m is expected to increase both surface and subsurface inflows to the channel. This would provide additional rearing habitat during the summer low flow period. It might also reduce the potential for fish to become trapped in these habitats over the summer. This site has the advantage of easy access from the adjacent road which would facilitate construction.

The main disadvantage with this site is that the excavation could infill over time as a result of sediment deposition. The proposed work area would, therefore, require monitoring and possible maintenance.

6.2.3 Site C. – Existing High Water Back Channel.

This back channel is located downstream of the sewer line crossing on lower Mayo River (Figure 6.2.1). The lack of permanent vegetation indicates that the channel is back flooded during periods of high flow. This type of habitat is likely used by fish as a refuge from mainstem freshet flows. The upper portion of the channel crosses an access road near the sewer lagoons. There were signs of seasonal flow near the road crossing.



Photo 8. Downstream view of high-water back channel area.



Photo 9. Upstream view of back channel taken from confluence with Mayo mainstem.

It would be easy to excavate this back channel as there is a nearby road and work could be undertaken in the 'dry' during the late summer low flow period. Deepening this channel could facilitate fish access into this area during a wider range of water levels in Mayo River. While it is possible that excavation of this area could recruit groundwater, it is anticipated that water levels would only be sufficient to create back channel type habitat. Over the longer term, the deposition of bed material near the confluence with Mayo River could cause this channel to become isolated during low flow. This could necessitate periodic re-excavation work to maintain a connection to the mainstem river. The other concern with this site is its proximity to the sewer lagoons, which work on the principle of evaporation/exfiltration. There is potential for capturing groundwater affected by the lagoons.

6.2.4 Site D – Former Gravel Pit

This excavated pond is located between the left bank dyke protecting the town of Mayo and the mainstem channel. (Figure 6.2.1). This former gravel pit is connected to the river during moderate to high flows. The site was accessible by fish on Sept 30th and isolated July 27th (Photos 10 and 11). This pond has similar habitat to a back channel and would provide refuge type habitat during periods of high flow. Cover is generally limited to the margins in the form of overhanging vegetation and woody debris. Use of this area by juvenile chinook may, therefore, be limited by predation from northern pike. One juvenile chinook was captured on September 30th. No fish were captured during low flows on July 27th.



Photo 10. Overview looking downstream in July.



Photo 11. Cross view of pond in September.

Excavation of this pond would facilitate full time fish access as well as possibly improved rearing habitat by increasing the depth. This could provide good habitat for emergent (0+) chinook, as they often utilize slow moving water until they are about 50mm (von Finster, Pers. comm. 2004). This site is also easily accessible and construction could be undertaken “in the dry” during periods of low flow.

The main disadvantage is that there is some uncertainty as to whether the proposed work would benefit chinook salmon. The lack of cover and flow as well as unrestricted access from the mainstem may result in predation by northern pike.

6.2.5 Site E – Existing Groundwater Channel.

This groundwater fed channel is located upstream of the highway crossing of the Mayo River (Figure 6.2.2). Measurements on July 27 indicate that emerging ground water is flowing at a temperature of 2.5°C. The channel extends 250 m before entering a side channel of the Mayo River (Photos 12 and 13). The upper 200 m is shallow (10-20 cm) and flows through a mix of organics, gravel and fine sediments. There is little habitat complexity or cover within the channel. The lower 50 m is deeper and is influenced during moderate flows by water levels in the mainstem Mayo. The entire channel would likely be back flooded during periods of high water. Juvenile chinook were captured in the lower 50 m of this channel in October. As there is a substantial amount of flow in this channel, there is potential to create significant rearing habitat by making it deeper and adding complexity consisting of pools, woody debris and boulders. Increasing the channel width would allow the water to warm up during the summer creating more optimum rearing temperatures. A disadvantage of this site is that it is located, approximately 500 m from the nearest road. This will require a trail to be constructed over comparatively easy ground. Existing cut-lines would, however, facilitate machinery reaching this area

with minimal vegetation disturbance. Groundwater flow is expected to occur for most of the year and sediment control measures would therefore be needed during the excavation period.



Photo 12. Upstream view of existing groundwater channel.



Photo 13. Downstream view of groundwater channel.

7.0 CONCLUSIONS

The apparent post-1949 loss of secondary channels in the area downstream of Wareham Dam is a significant concern for chinook rearing. Habitat restoration focused on increasing the rearing capacity of the Mayo River is therefore recommended to offset these losses.

A number of potential restoration sites were identified. Of these, the side channel (Site B) and the groundwater channel (Site E) have the highest potential for creating productive juvenile chinook habitat. These sites both appear to have enough water to provide sufficient flow and will function like a side channel or a tributary stream, both desirable rearing habitats for juvenile chinook. Deepening these channels would allow for improved access during most flow conditions. Of these two sites, Site B is the most desirable as it has easier access, would be easier to construct and likely has more favorable water temperatures. Site E, however, could be used to investigate the feasibility of enhancing cold groundwater fed channels.

The potential groundwater channel (Site A) might also be a lower priority candidate for restoration/improvement. However, additional investigation would be required to ensure that ground water volumes would be sufficient to maintain the desired habitat conditions.

The two back channels (sites C and D) both are easily accessible and would be very easy to construct. Of these two sites, Site D would be preferable due to Site C's proximity to the sewer lagoons. The deepening and monitoring of Site D would help quantify the value of creating back channel habitat for chinook. The literature and our sampling results indicate that chinook use of back channels is limited; therefore, creating these types of habitat may be a lower priority than areas with flowing water.

8.0 RECOMMENDATIONS

Once a decision has been made to proceed, designs and cost estimates should be prepared and environmental approvals will need to be obtained. The identified work area encompasses a variety of site conditions and provides opportunities to evaluate various techniques for restoring juvenile chinook habitat in the Yukon Territory. As such, an intensive monitoring program to document fish utilization throughout the year and post-construction physical changes to the works is recommended. The results can then be used to assess the value of these works as a method of restoring or improving juvenile chinook habitat.

The following recommendations should be followed.

1. An Environmental Management Plan should be completed to describe the works and outline the mitigation strategies (i.e. fish salvage and erosion/sediment control) to reduce construction impacts and optimize the design of the proposed works. This will help to address any regulatory agencies concerns and aid in the permitting process.
2. Works should be conducted during low flows to allow for easier construction and minimize the potential for disturbance.
3. Cover features such as woody debris and large rock (where available) should be incorporated into the design and construction of these channels.
4. In the case of the existing side channel (Site B), the lower 250 m of the side channel should be excavated, leaving the upper 50 m intact to minimize the potential for the main channel to shift into this side channel.

8.1 Physical Studies

The recent construction of a transmission line to Dawson will change the operating regime of Wareham Dam. This may influence the physical stability and biological use of the secondary channels in lower Mayo River. It would, therefore, be desirable to undertake a monitoring program to assess how regulation is affecting water levels, water quality and channel connectivity. It would also be useful to quantify the length of secondary channels which can be identified on the historical air photos and evaluate changes in channel number and length over time. Ideally this work should include field verification of present conditions and a more detailed analysis of how construction of the Wareham and Mayo Lake dams has affected the downstream hydrology.

8.2 Biological Studies

Although documented, the use of the Reach 5 for chinook spawning and incubation is not well understood. Further study is required to determine how many fish are spawning in this area and what the success rate of the eggs during the incubation period. It may be beneficial to physically prevent spawners from entering this area during the spawning season. Alternatively if stock restoration via incubation or re-establishment of chinook above the dam is deemed desirable, the adult chinook using this area could be used as a source.

9.0 LITERATURE CITED

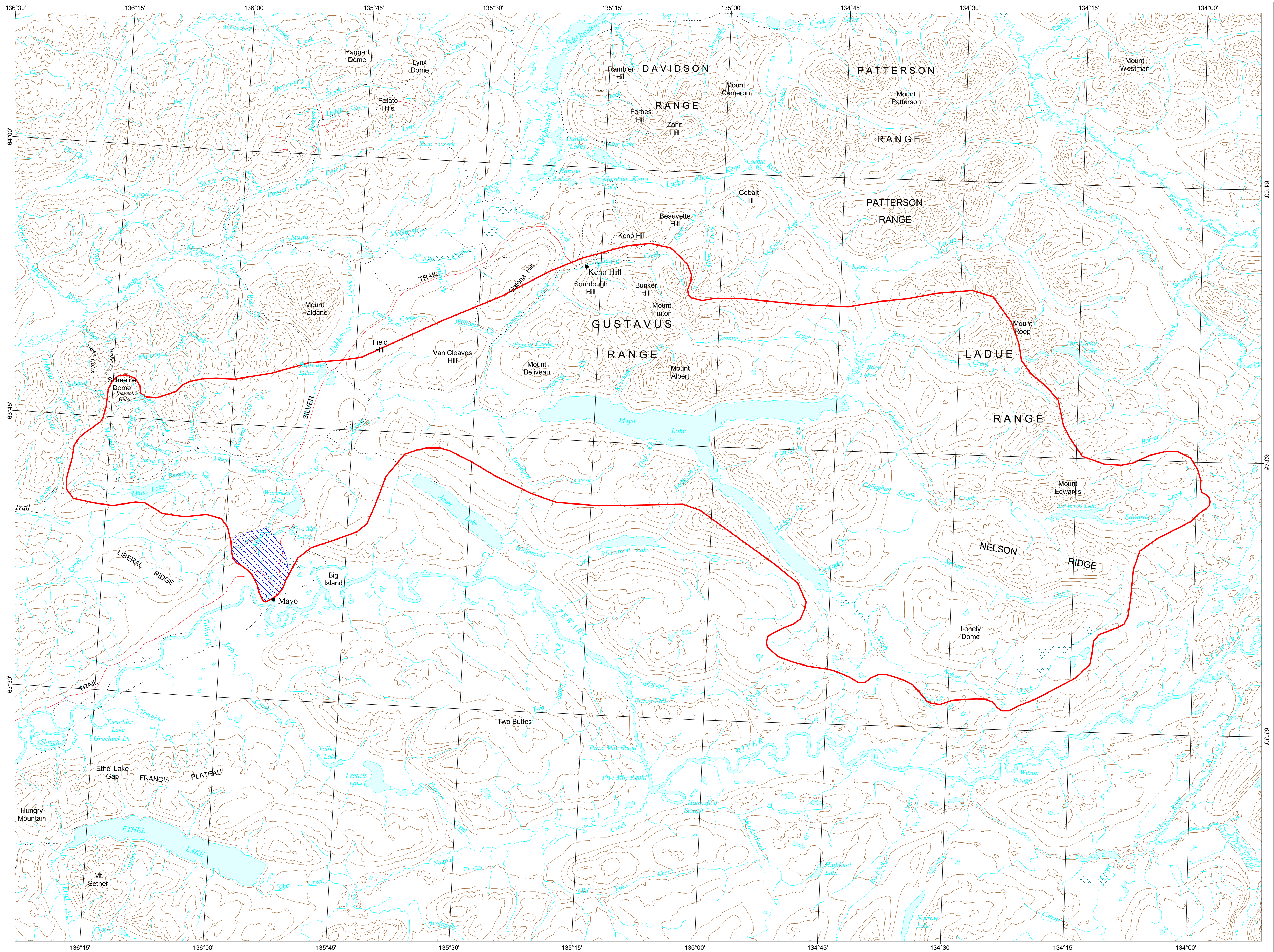
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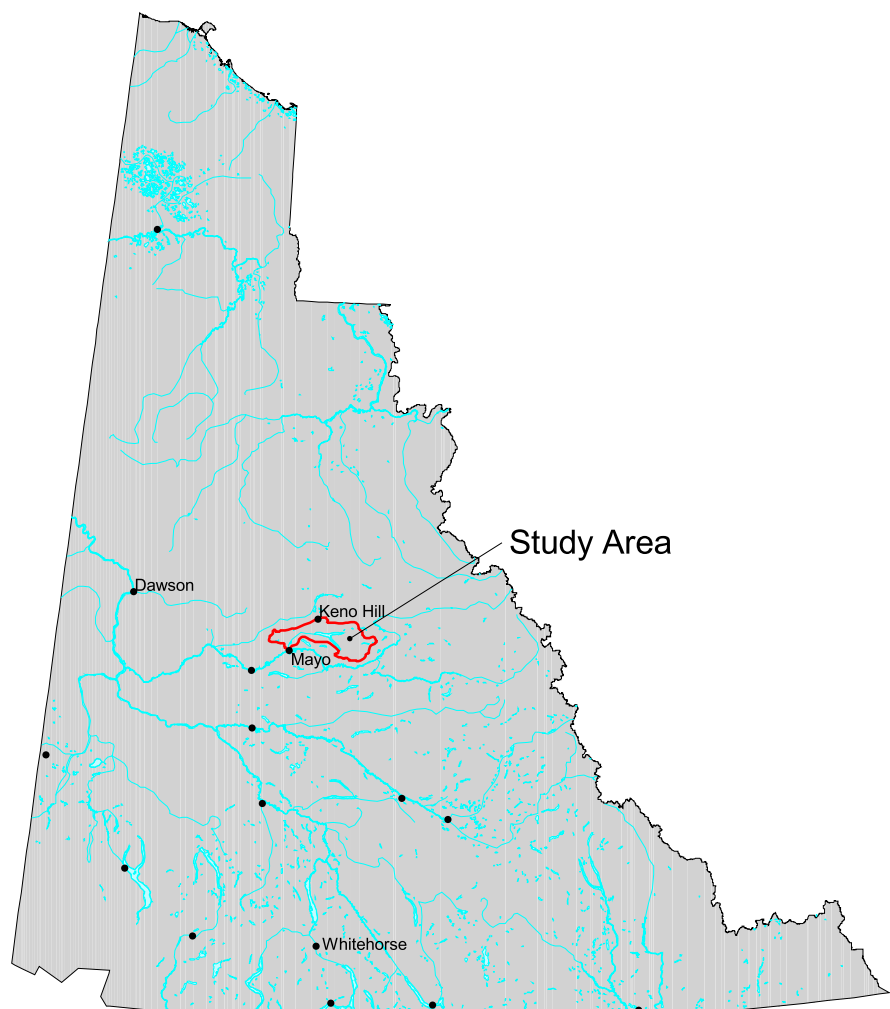
APPENDIX A: Figures



Mayo River Watershed

Scale 1:200,000
5 0 5 10 Kilometers
Albers Equal Area projection (standard parallels: 61°40'N & 68°00'N)

- Mayo River Watershed
- Study Area
- Streams
- Elevation
- Roads
 - LIMITED-USE ROAD
 - ROAD
 - TRAIL



Location of the Mayo River Watershed within the Yukon

DIGITAL DATA SOURCES AND DISCLAIMERS

National Topographic Data Base (NTDB) compiled by Natural Resources Canada at 1:250,000. Reproduced under license from Her Majesty the Queen in Right of Canada with permission of Natural Resources Canada.

Map Compiled By:
Millie Olsen
Lands & Resources Department
Nacho Nyak Dun First Nation
Mayo, Yukon
30 October 2003



Figure 2.1 Location of study area within the Mayo River Watershed

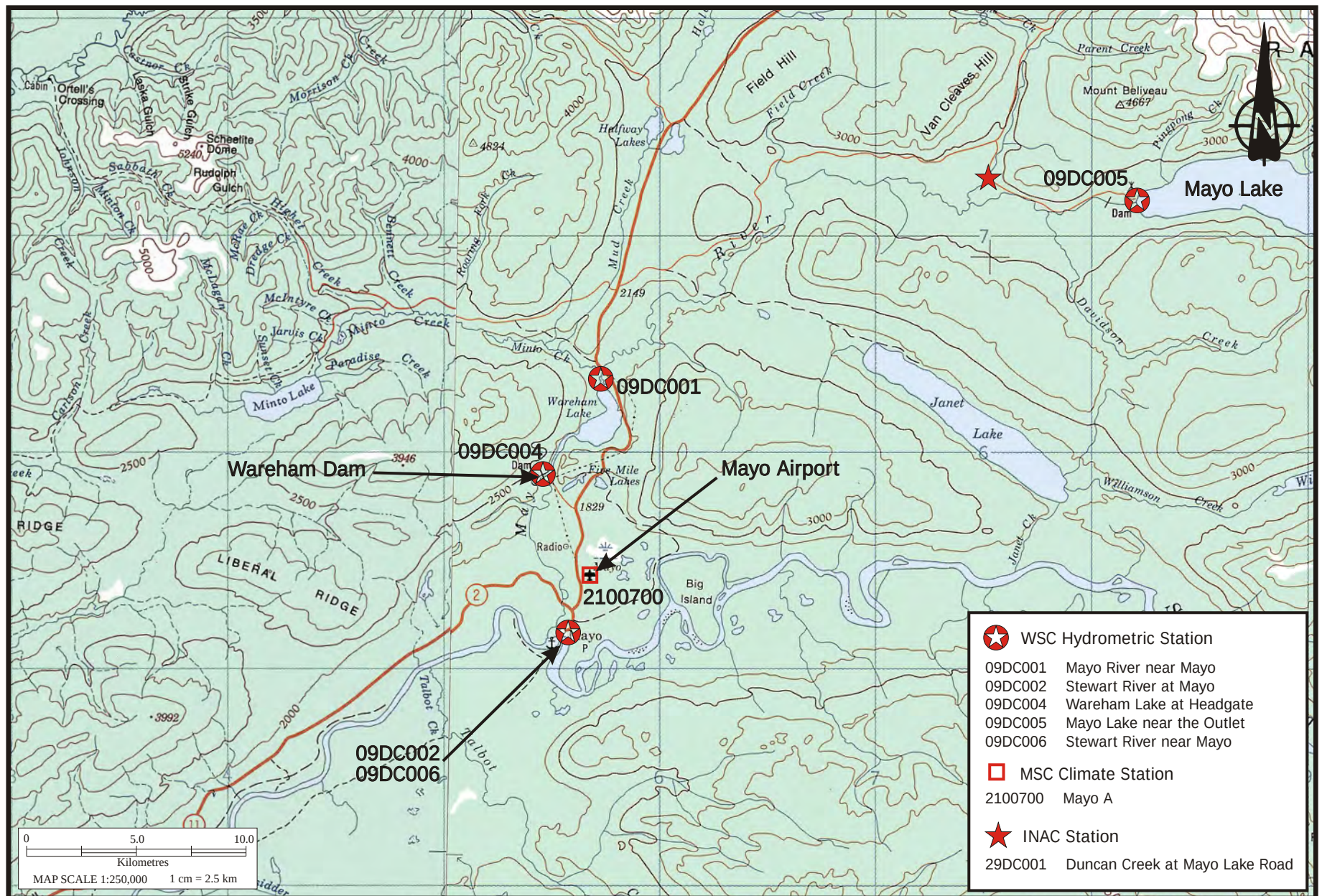


Figure 4.1.1: Map showing location of hydrometric and climate stations.

MAYO AIRPORT - NORMALS 1971-2000

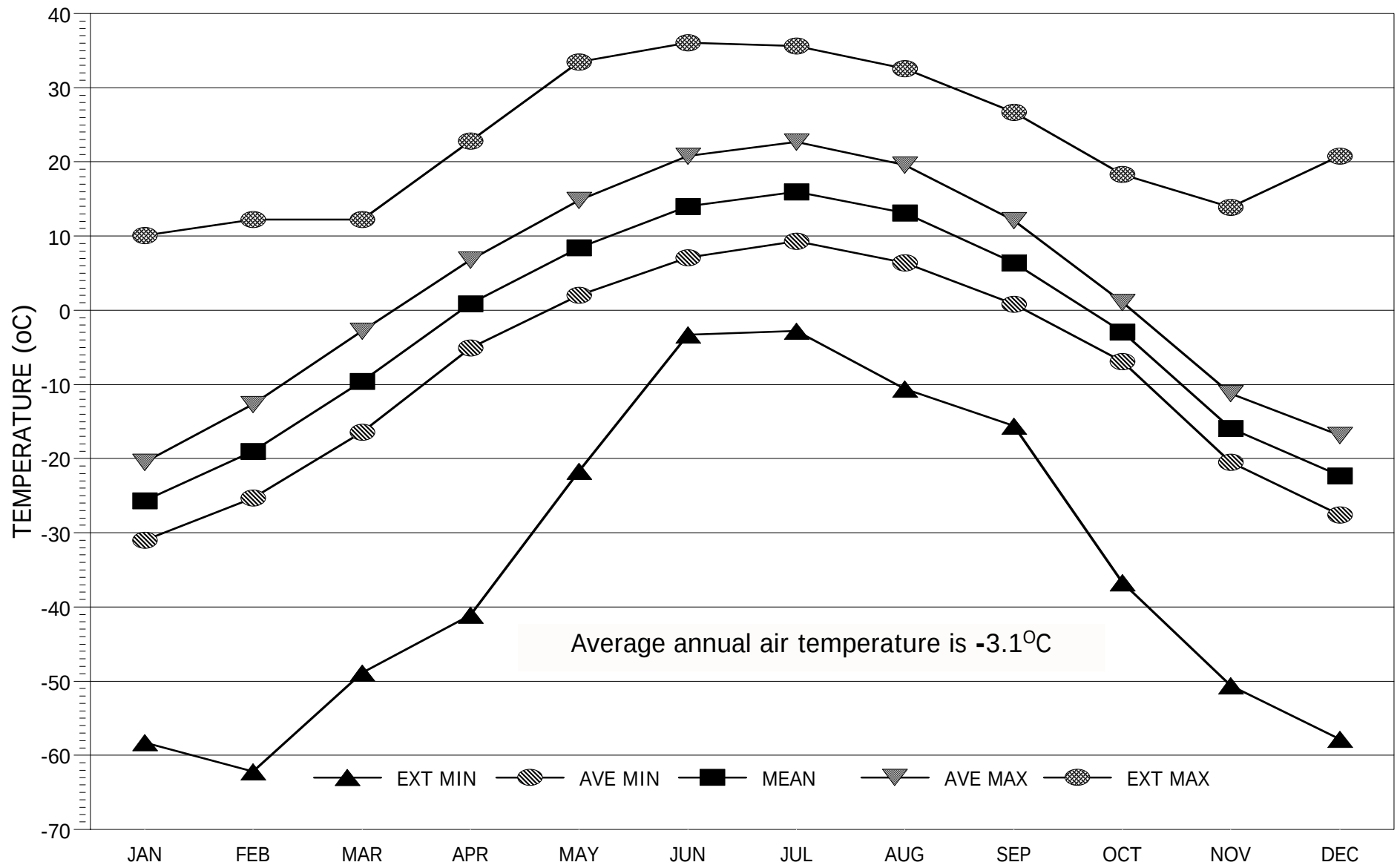


Figure 4.1.2: Seasonal variation in temperature at Mayo Airport.

MAYO AIRPORT - NORMALS 1971-2000

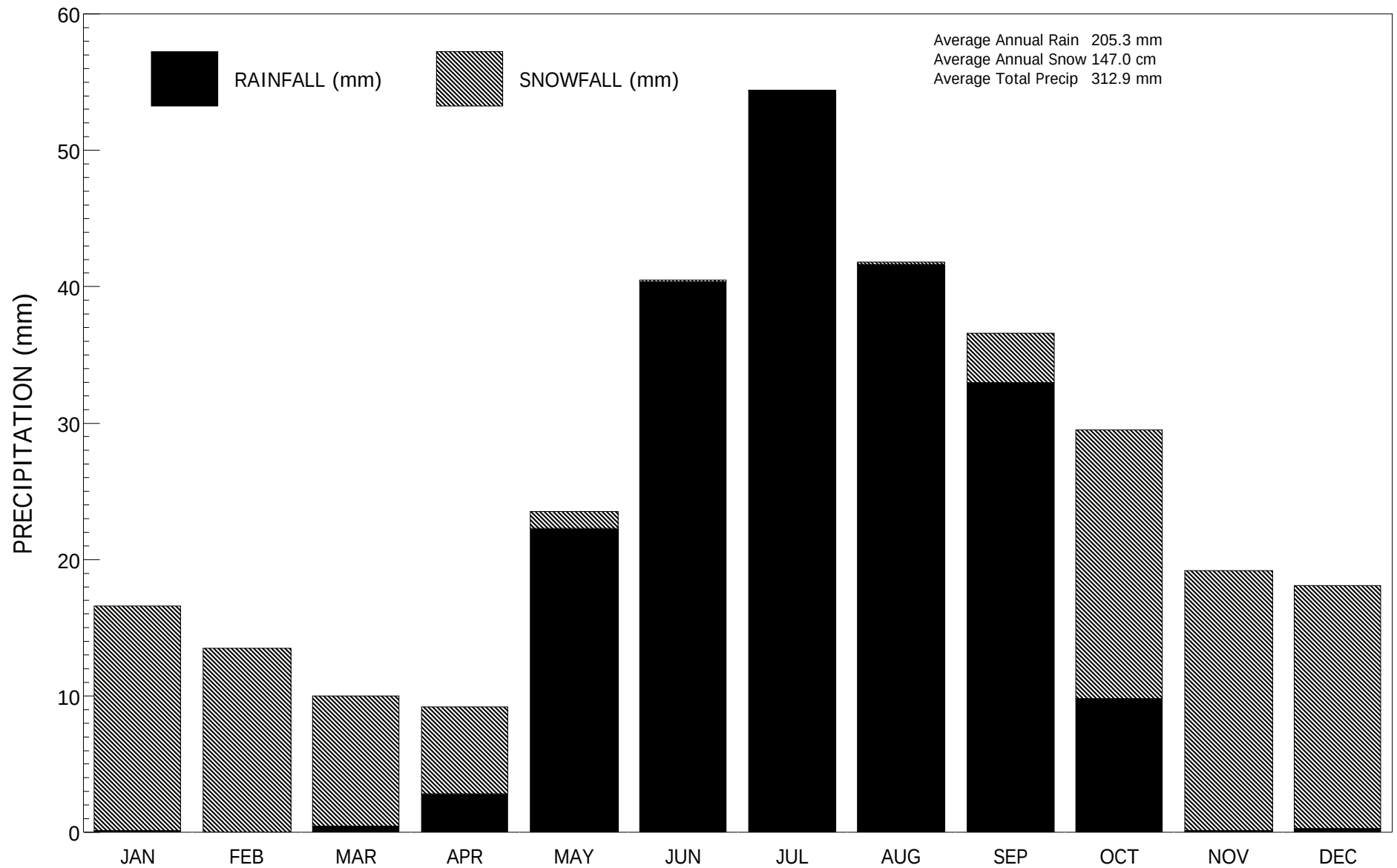


Figure 4.1.3: Seasonal variation in mean monthly precipitation at Mayo Airport.

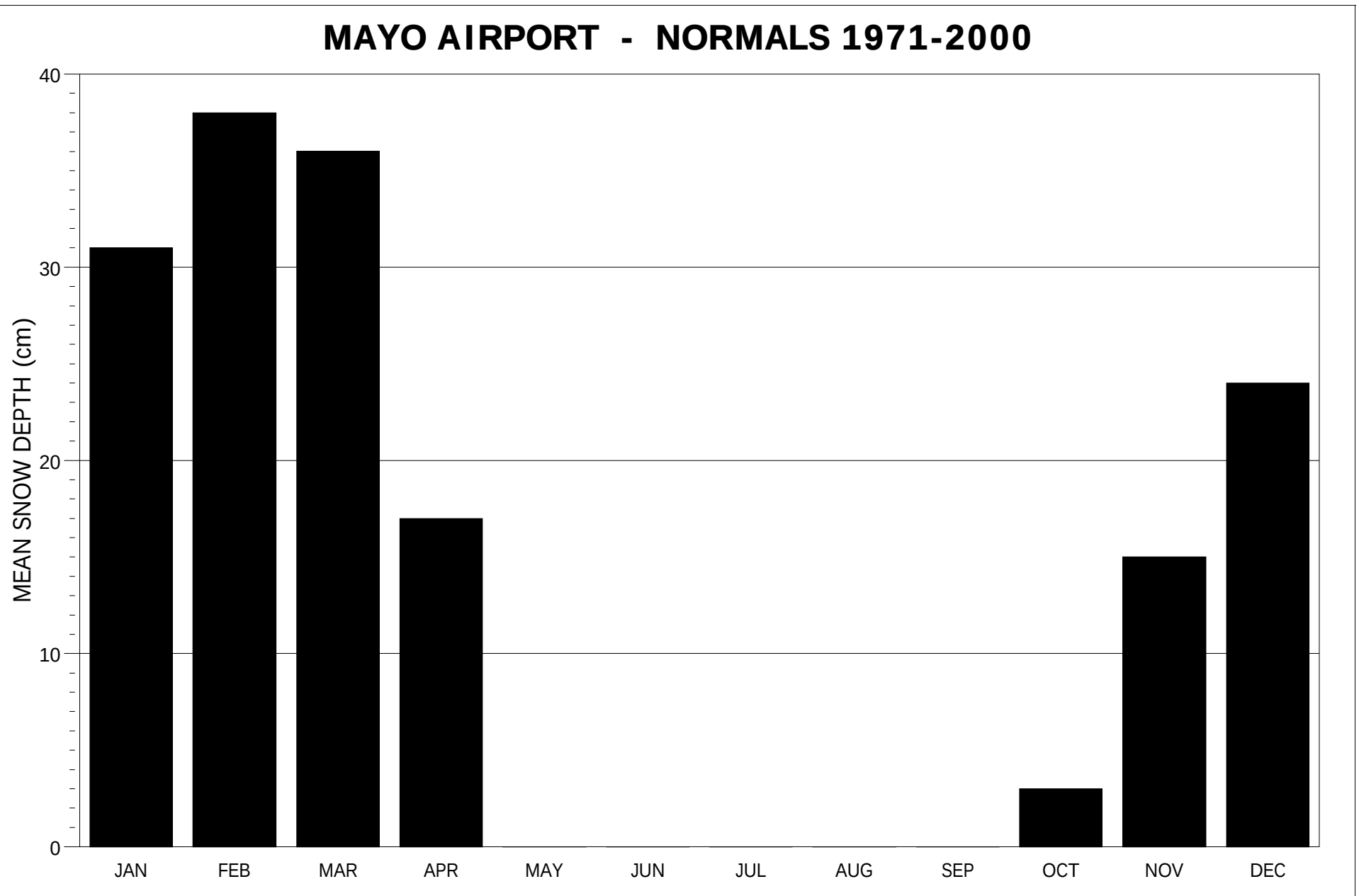


Figure 4.1.4: Seasonal variation in mean snow depth at Mayo Airport.

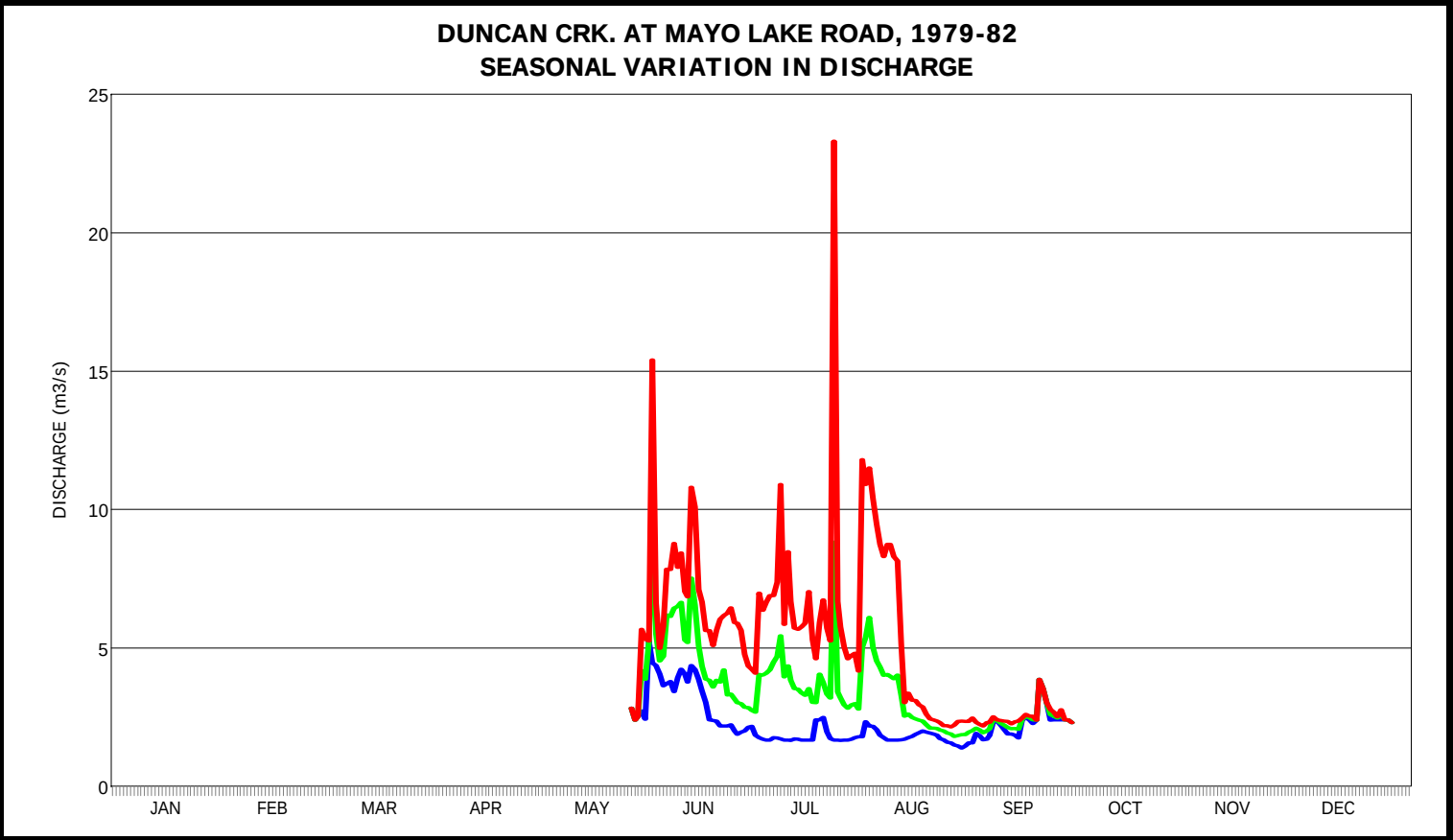
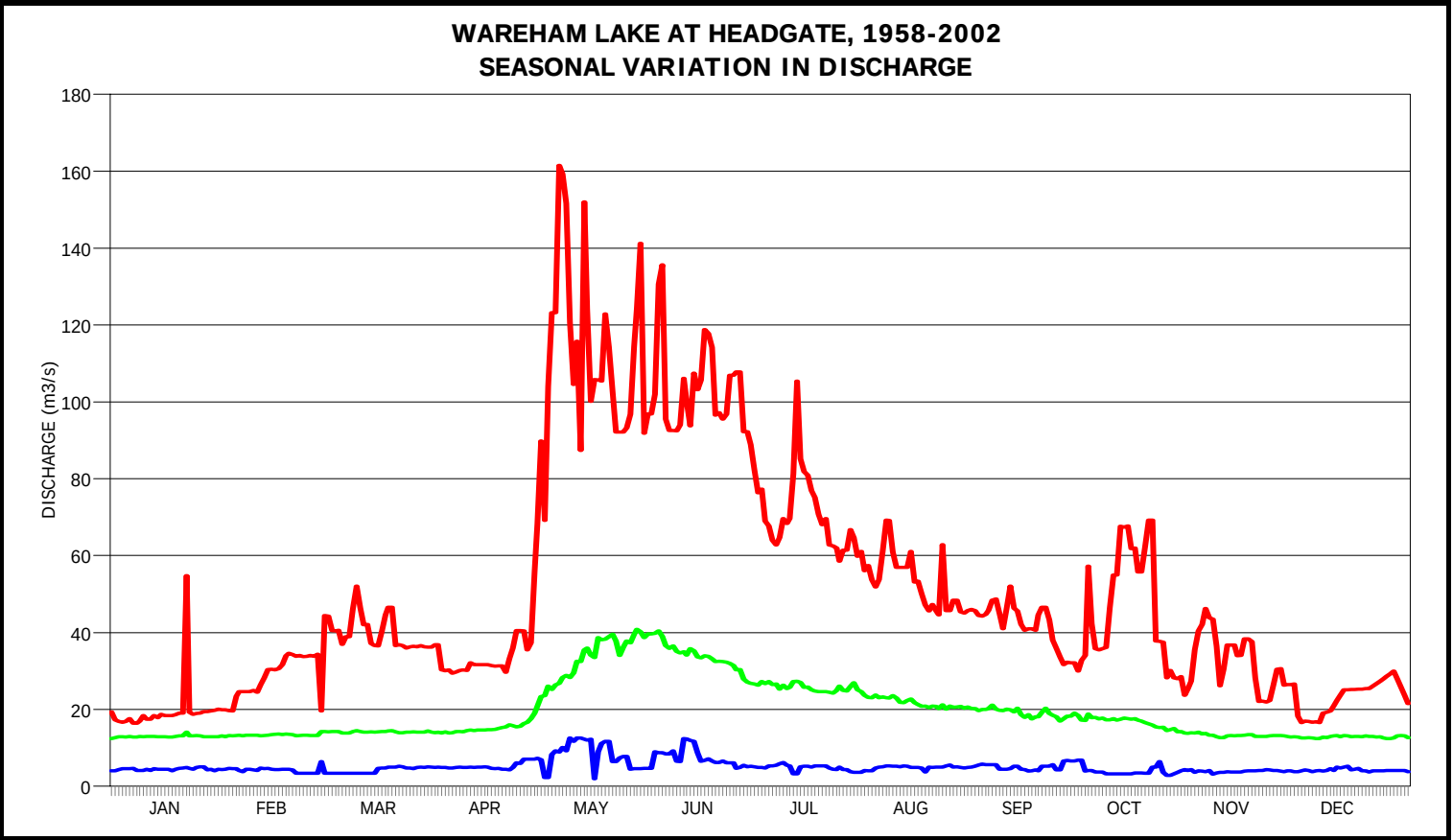
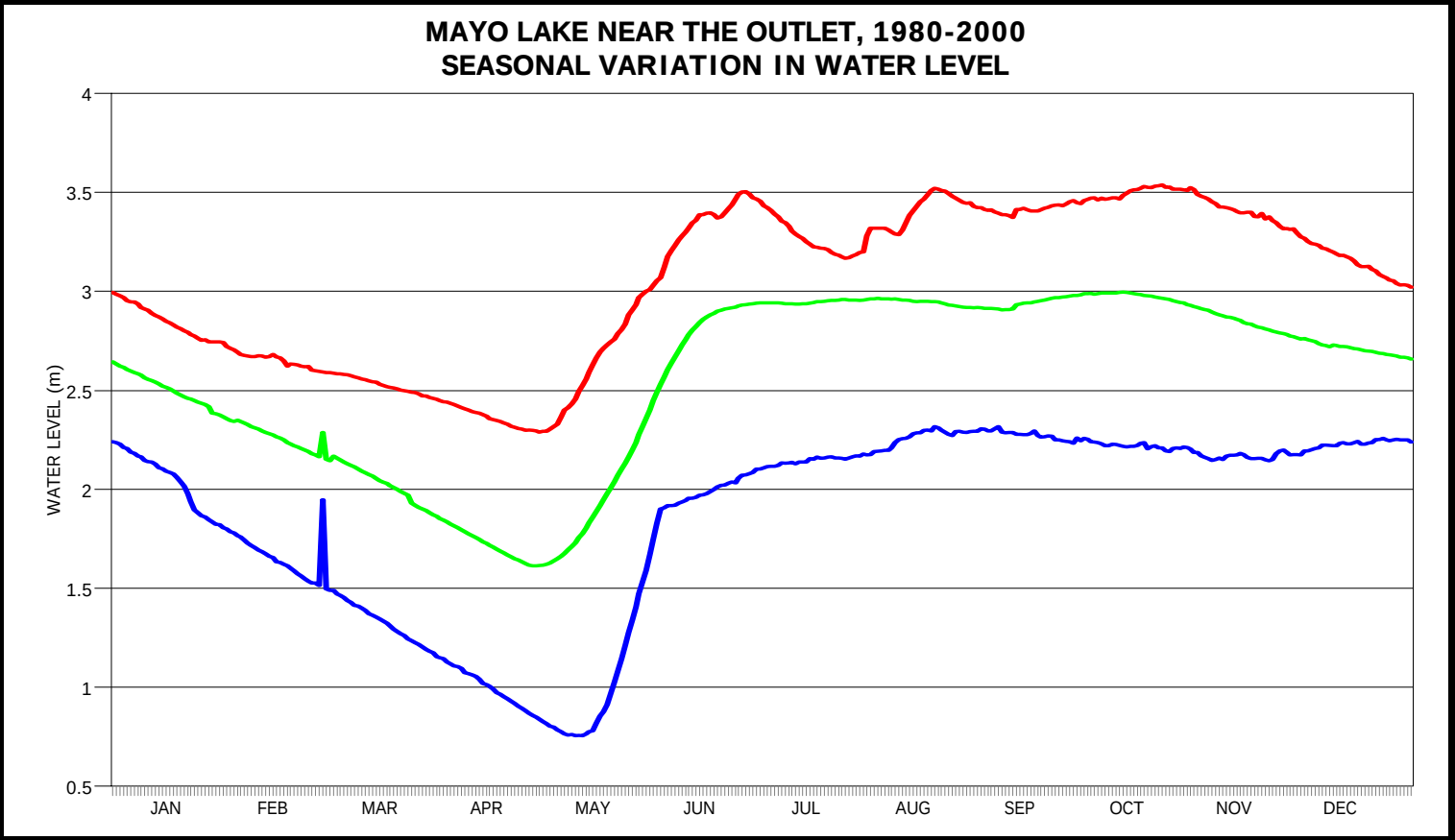
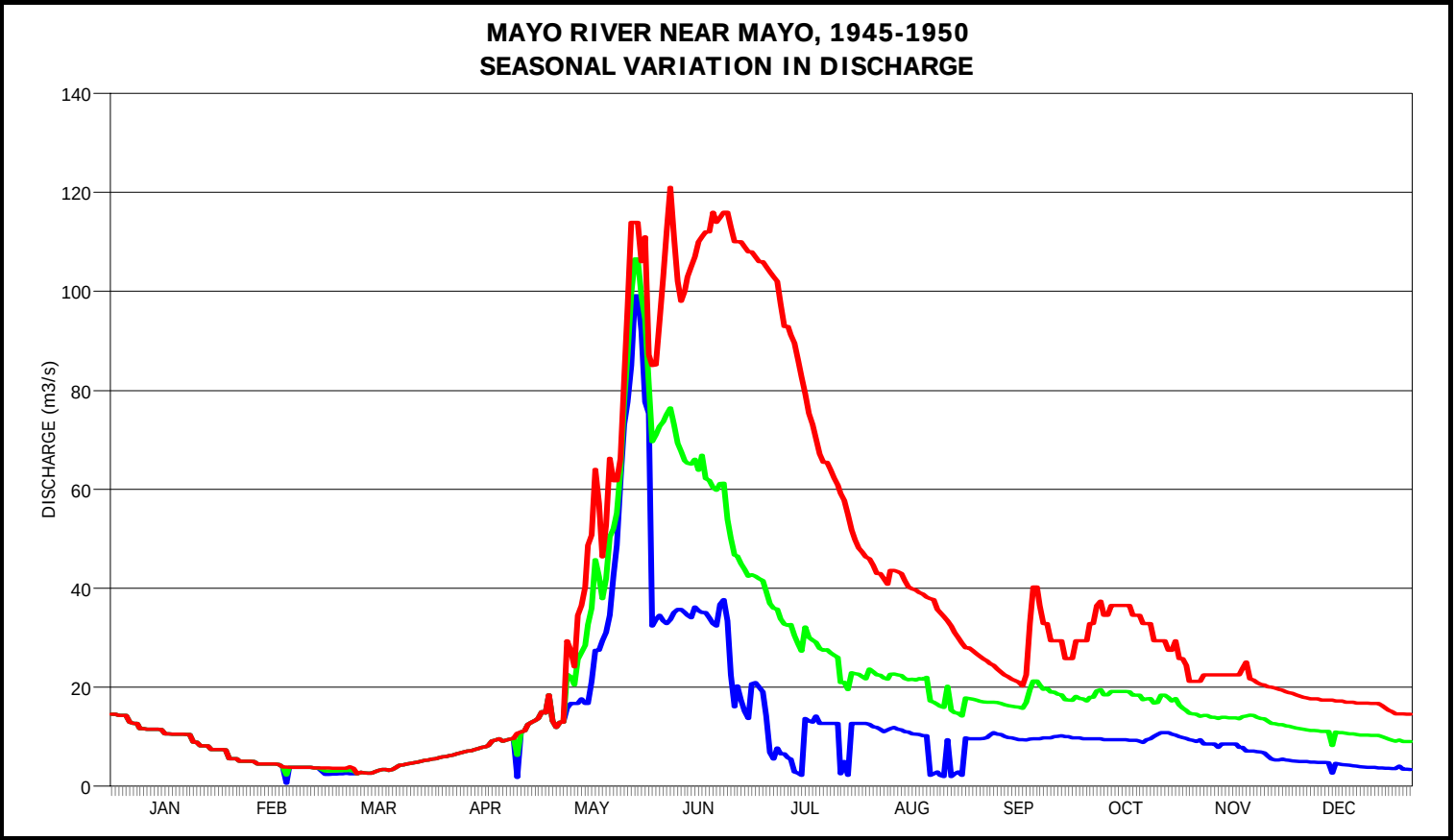


Figure 4.2.1: Seasonal variation in discharge or water level.

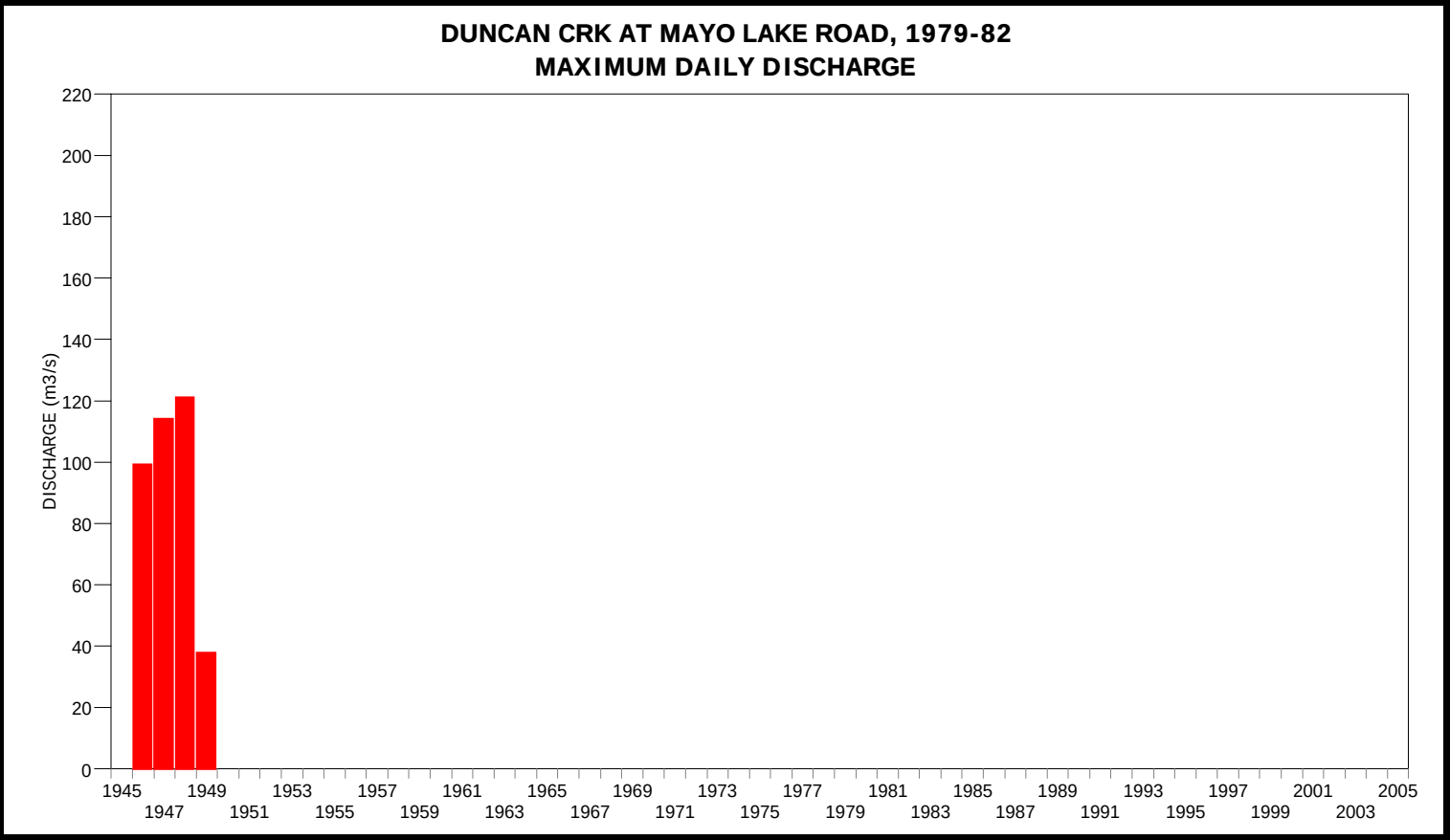
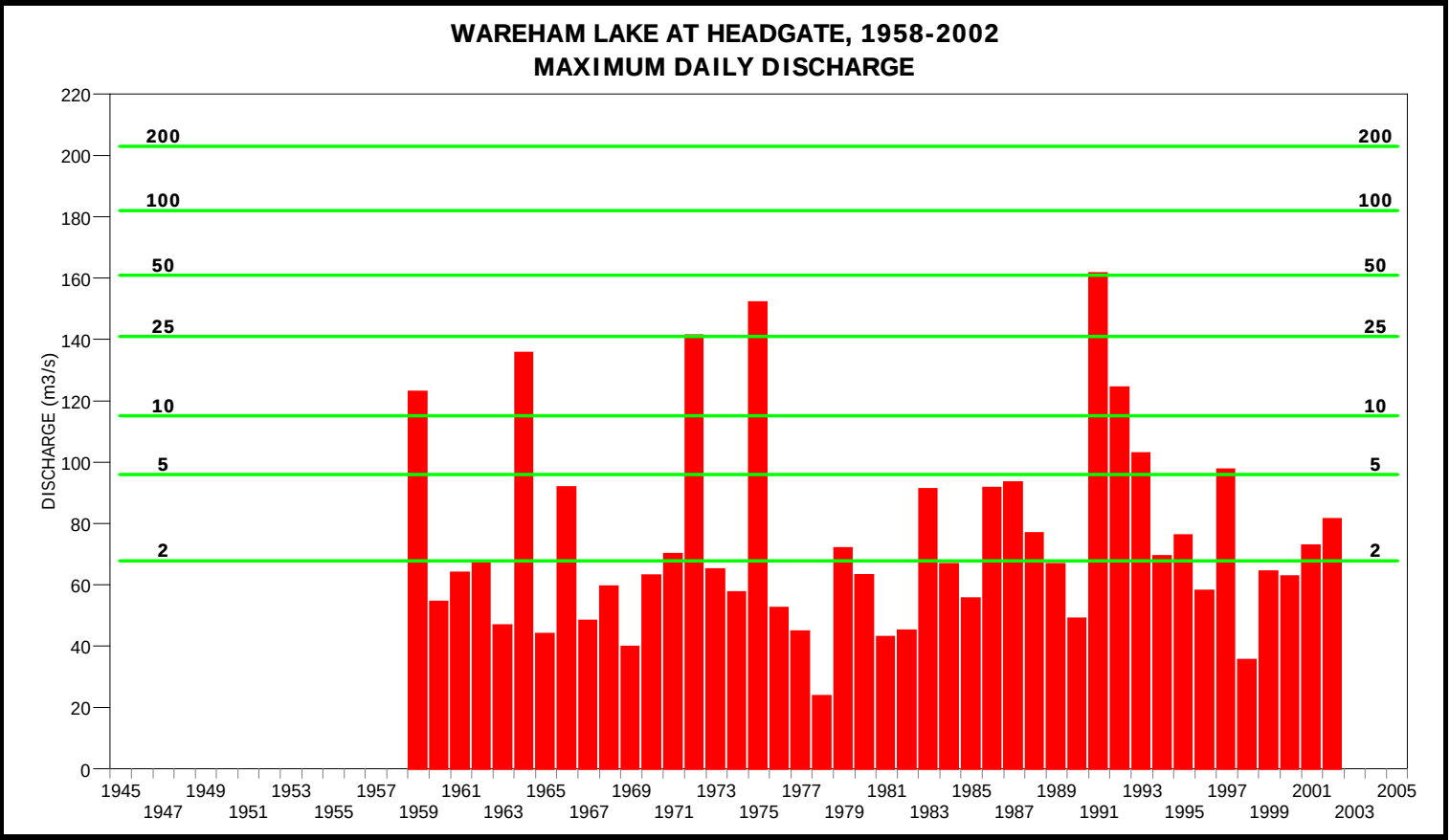
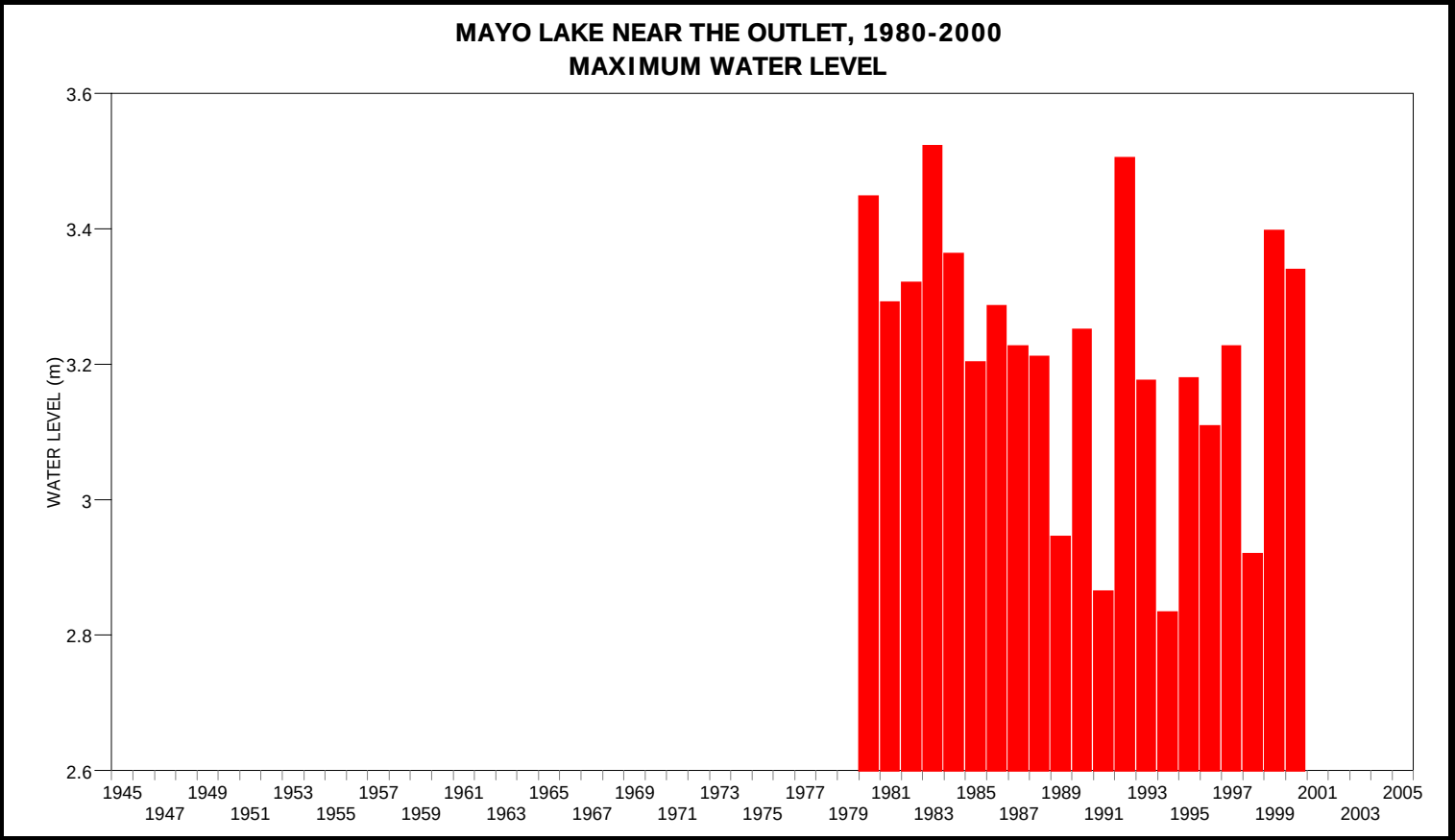
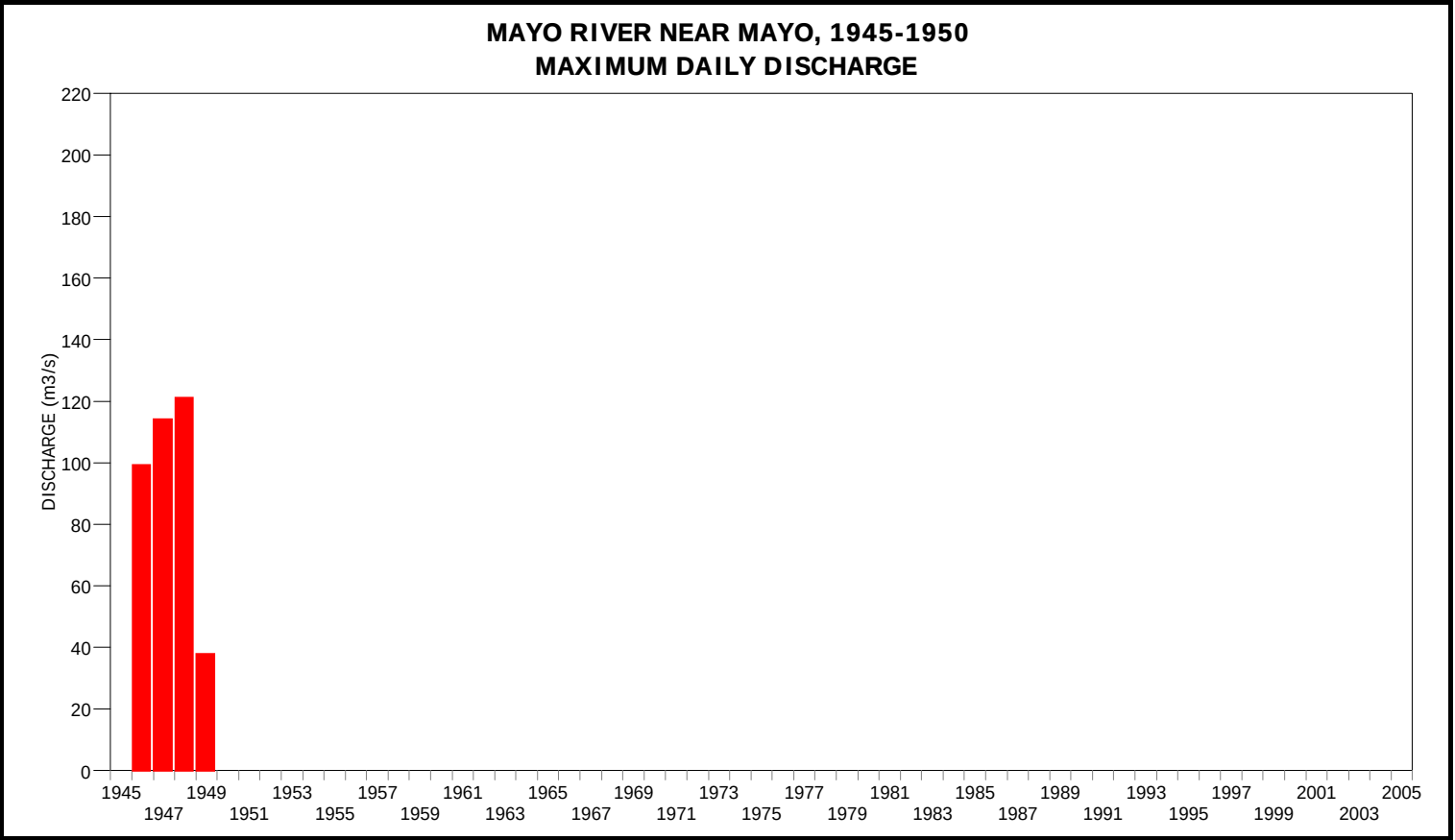


Figure 4.2.2: Historical variation in annual maximum daily discharge or water level.

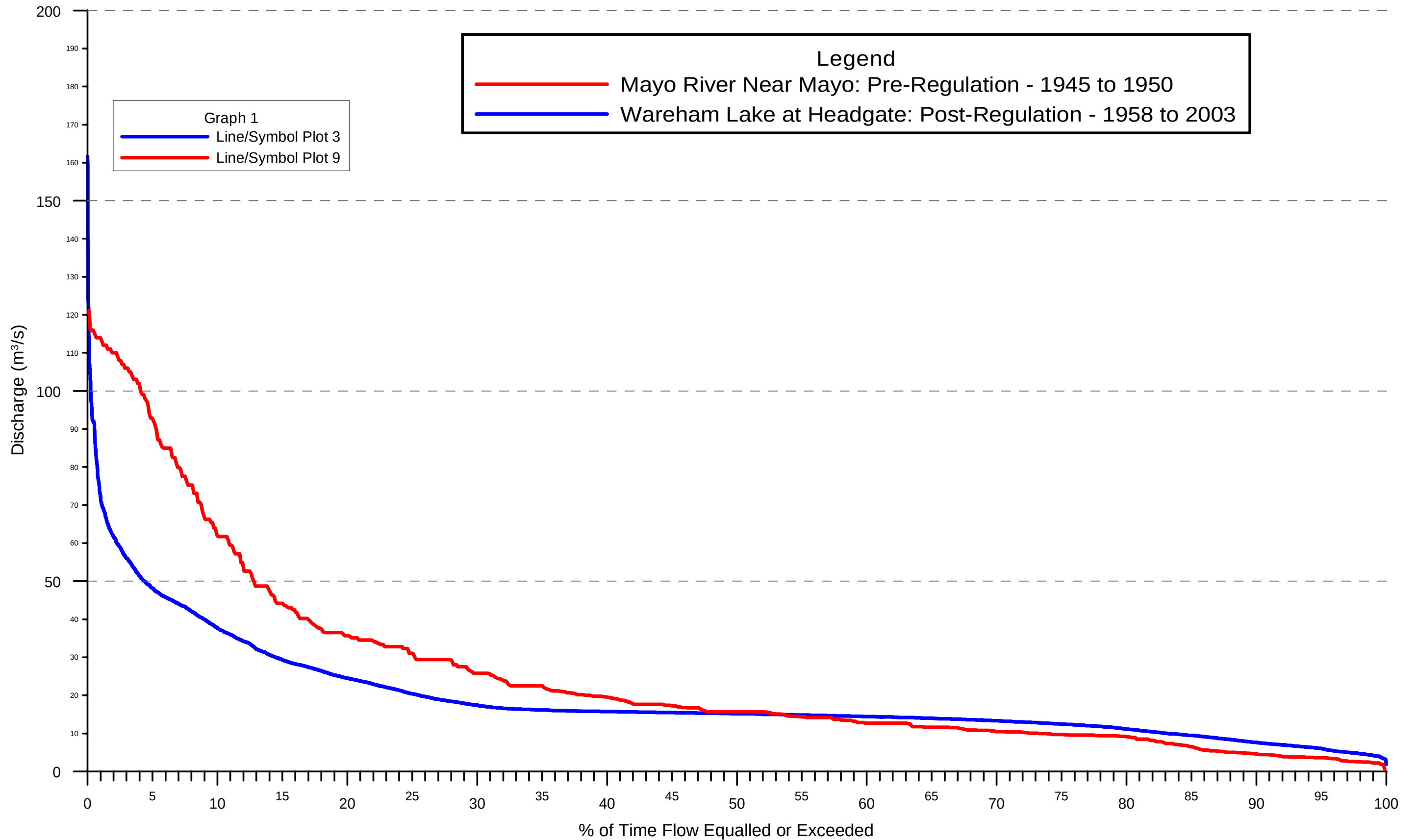


Figure 4.2.3: Annual flow duration curve for natural and regulated flow regimes.

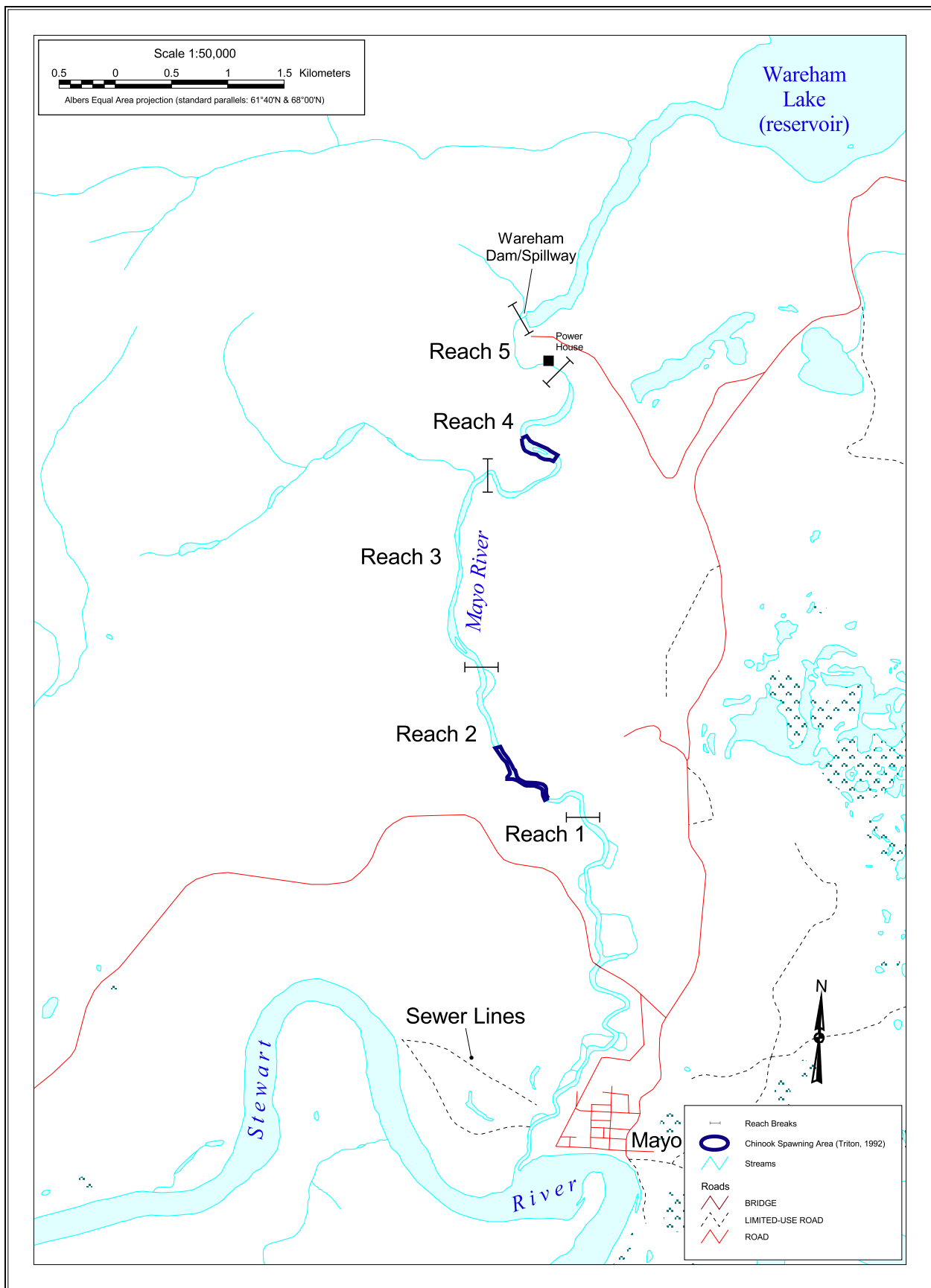
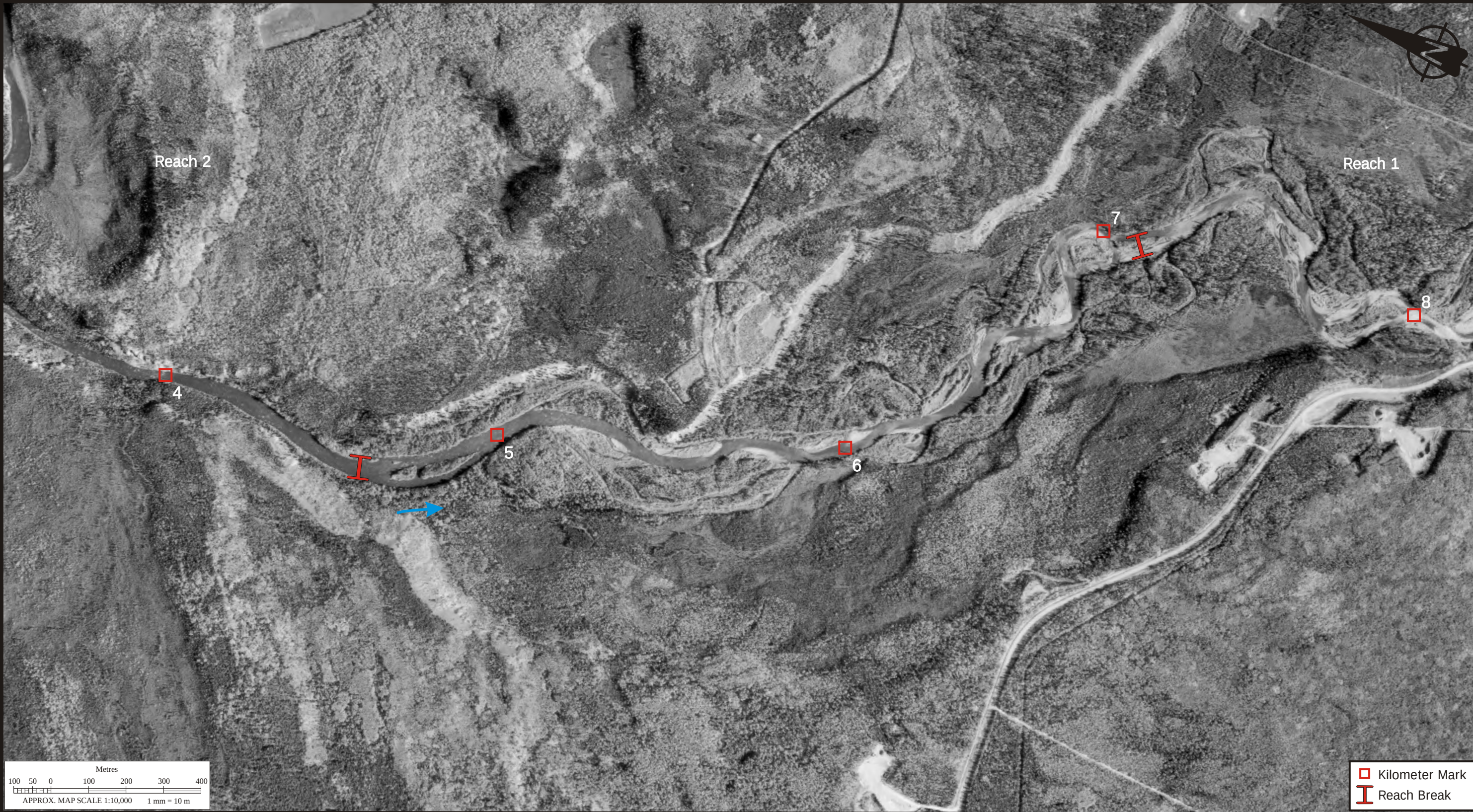


Figure 4.3.1 Location of reach breaks and spawning areas in the lower Mayo River.



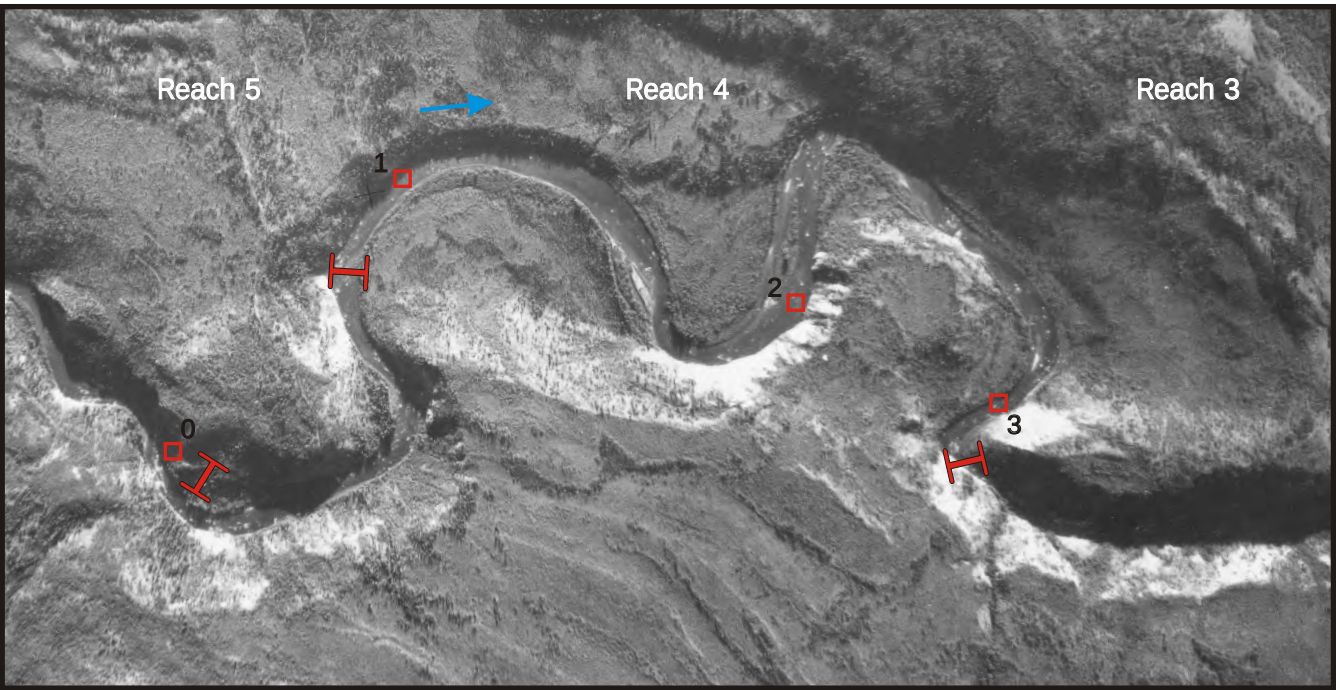
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						DATE: December 16, 2003				TITLE: 1996 AIR PHOTO MOSAIC OF REACHES 5, 4, 3 AND 2		
REFERENCED DRAWING NO.			REFERENCED DRAWING DESCRIPTION			DRAWN: S. Allegretto						
A	July 8, 2003	Issued For Discussion		SA	MM		DESIGNED: S. Allegretto			CLIENT: YUKON RIVER PANEL BOX 20973 WHITEHORSE, YUKON, Y1A 7A2		
B	Dec. 16, 2003	Issued For Report					CHECKED: M. Miles					
							APPROVED:					
									FIGURE 4.3.2		PROJECT NO. 240	REV: B
											Km NO. 0 to 5	



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						DATE: December 16, 2003				TITLE: 1996 AIR PHOTO MOSAIC OF REACHES 2 AND 1					
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A	July 8, 2003	Issued For Discussion		SA	MM		DESIGNED: S. Allegretto			CLIENT: YUKON RIVER PANEL BOX 20973 WHITEHORSE, YUKON, Y1A 7A2	FIGURE 4.3.3		PROJECT NO. 240		REV: B
B	Dec. 16, 2003	Issued For Report					CHECKED: M. Miles						Km NO. 4 to 8		
							APPROVED:								



A28299 #221			September 6, 1996 Air Photo			APPROXIMATE SCALE: 1:10,000			M. MILES AND ASSOCIATES LTD. 645 ISLAND ROAD, VICTORIA, BC, V8S 2T7 Phone: 250-595-0653 Fax: 250-595-7367 email: mmaa@coastnet.com CLIENT: YUKON RIVER PANEL BOX 20973 WHITEHORSE, YUKON, Y1A 7A2	PROJECT: MAYO RIVER		
						DATE: December 16, 2003				TITLE: 1996 AIR PHOTO MOSAIC OF REACH 1		
REFERENCED DRAWING NO.		REFERENCED DRAWING DESCRIPTION				DRAWN: S. Allegretto						
A	July 8, 2003	Issued For Discussion		SA	MM	DESIGNED: S. Allegretto						
B	Dec. 16, 2003	Issued For Report				CHECKED: M. Miles						
						APPROVED:			FIGURE 4.3.4	PROJECT NO. 240	REV: B	
										Km NO. 8 to 12		



(i)
Date: July 27, 1949
A12133 #187

NOTE:

- Pre-regulation (1952) channel conditions.

Discharge:	
Mayo River near Mayo	51.8 m³/s
Mayo Lake near the Outlet	na
Wareham Lake at Headgate	na

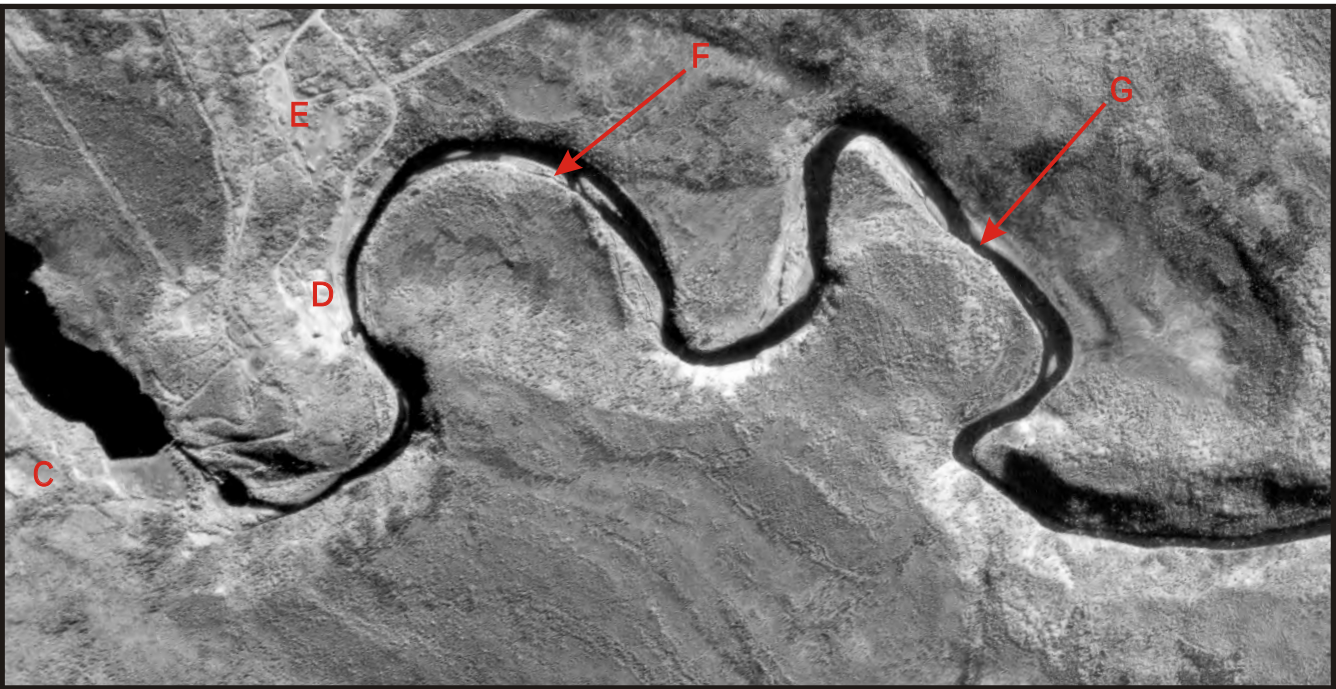


(ii)
Date: July 21, 1966
A19604 #26

NOTE:

- Wareham Dam (**A**).
- Powerhouse and Penstock outlet (**B**).
- Widespread construction related slope disturbance (eg. **C**, **D** & **E**).
- Increased size of unvegetated instream gravel accumulations in the vicinity of **F** & **G**.

Discharge:	
Mayo River near Mayo	na
Mayo Lake near the Outlet	na
Wareham Lake at Headgate	na

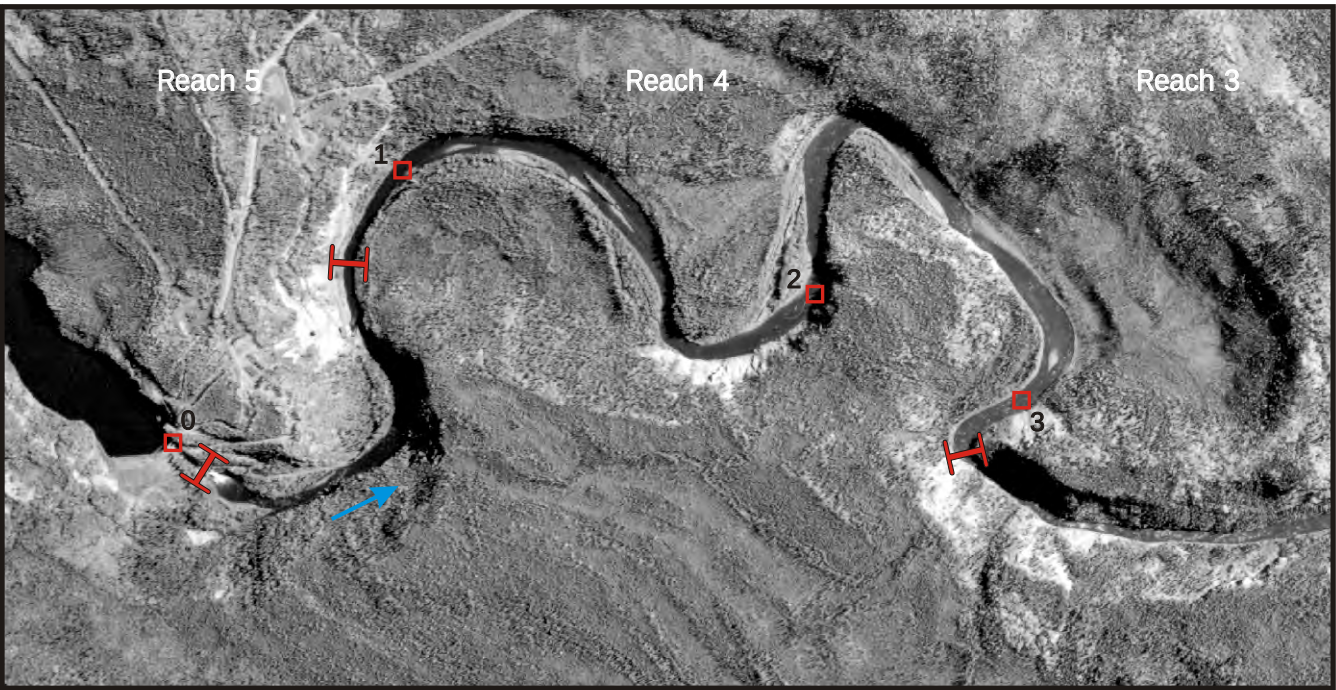


(iii)
Date: August 8, 1984
A26585 #152

NOTE:

- Widespread revegetation of disturbed slope (eg. **C**, **D** & **E**).
- Vegetation establishment on formerly unvegetated gravel bars in the vicinity of **F** & **G**.

Discharge:	
Mayo River near Mayo	na
Mayo Lake near the Outlet	2.998 m
Wareham Lake at Headgate	14.3 m³/s



(iv)
Date: September 6, 1996
A28299 #243

NOTE:

- Continued vegetation development on upland and riparian surfaces.
- Generally stable channel conditions with few unvegetated gravel bars.

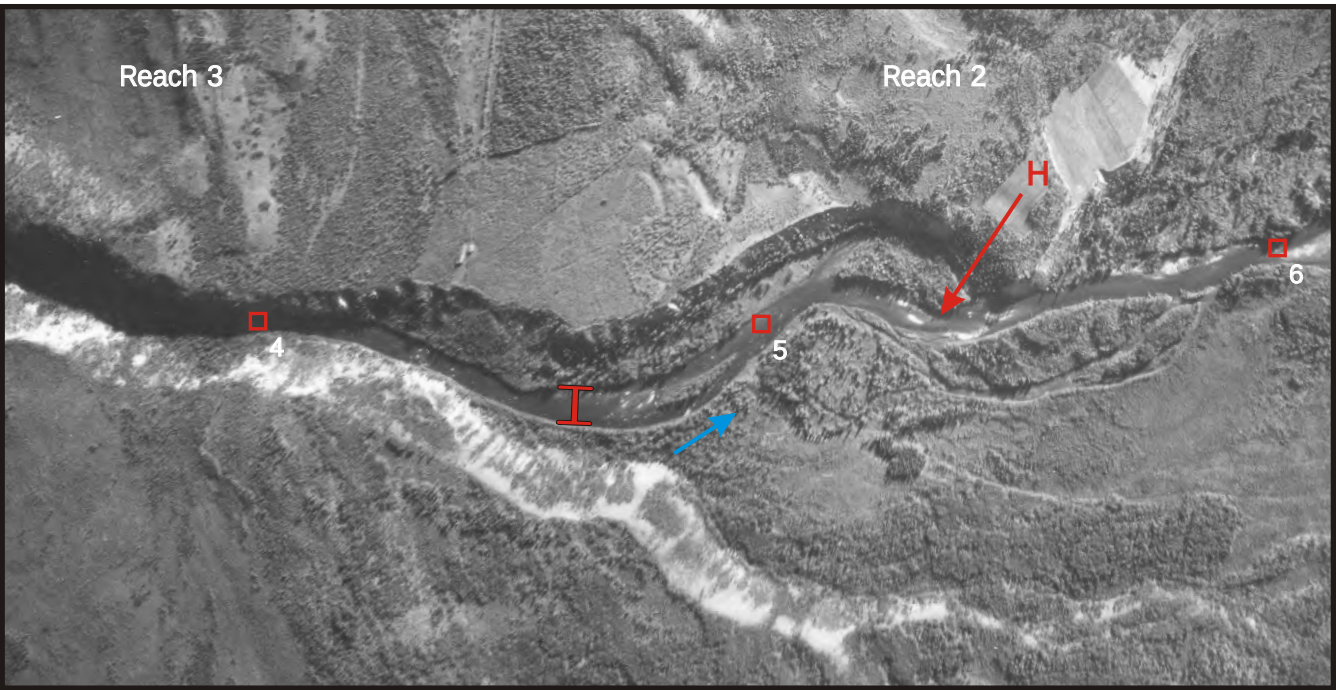
Discharge:	
Mayo River near Mayo	na
Mayo Lake near the Outlet	3.002 m
Wareham Lake at Headgate	19.0 m³/s

■ Kilometer Mark

I Reach Break

Figure 4.3.5: Historical changes in channel morphology (**reaches 5, 4 and 3**), Mayo River.

M. MILES AND ASSOCIATES LTD.



(i)
Date: July 27, 1949
A12133 #184

NOTE:

- Pre-regulation (1952) channel conditions.
- Small unvegetated instream gravel accumulations in the vicinity of **H**.

Discharge:

Mayo River near Mayo	51.8 m³/s
Mayo Lake near the Outlet	na
Wareham Lake at Headgate	na



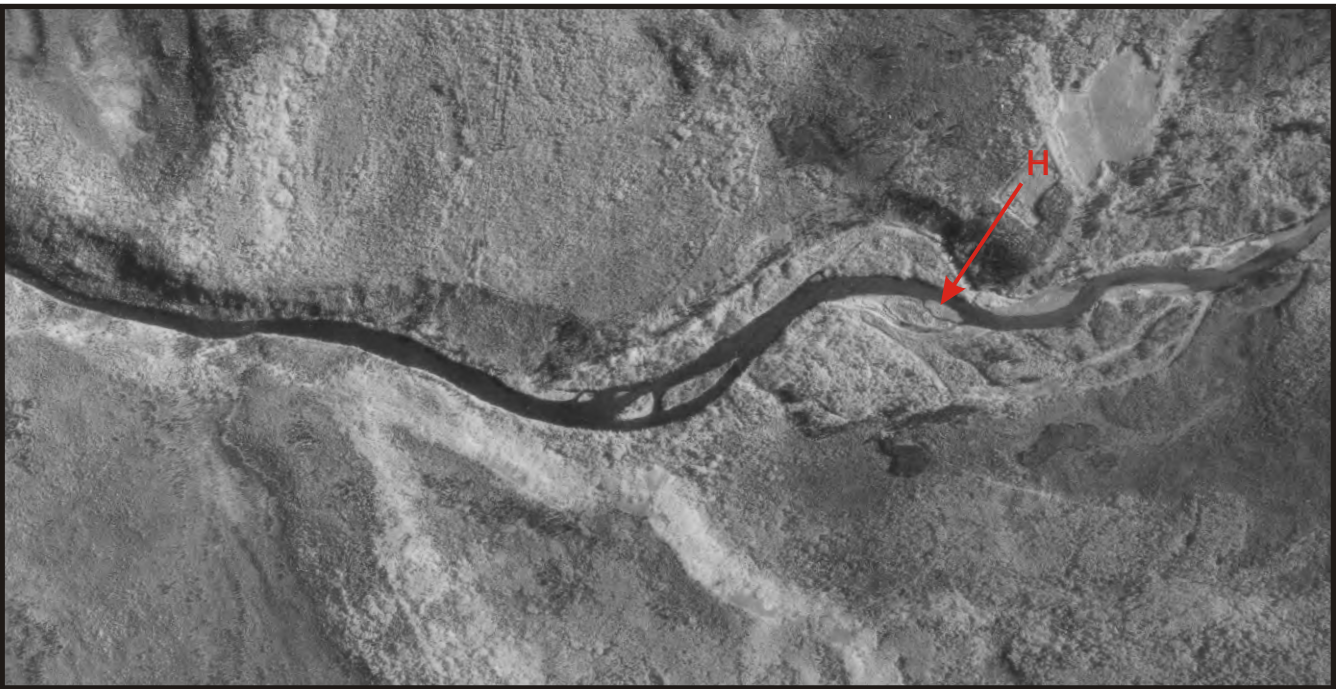
(ii)
Date: July 21, 1966
A19604 #26

NOTE:

- Post dam construction enlargement on unvegetated, instream gravel accumulations in the vicinity of **H**.

Discharge:

Mayo River near Mayo	na
Mayo Lake near the Outlet	na
Wareham Lake at Headgate	na



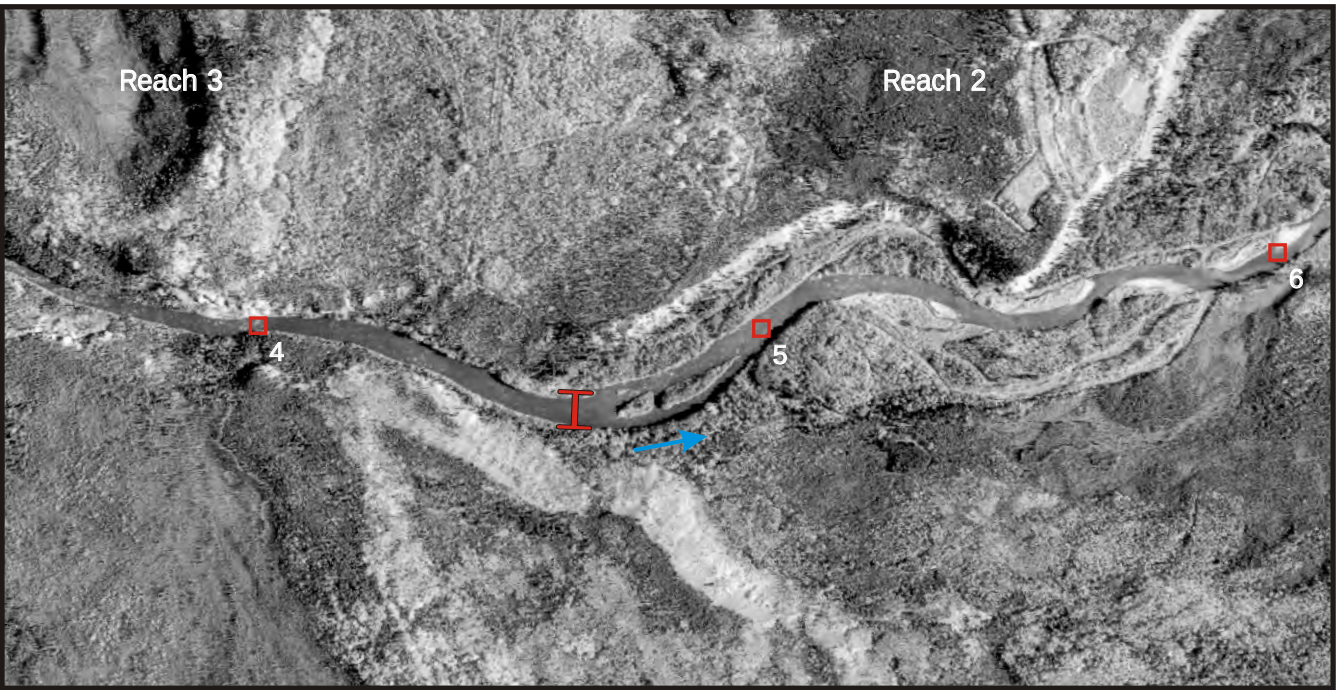
(iii)
Date: August 8, 1984
A26585 #152

NOTE:

- Vegetation development on formerly unvegetated, instream gravel bars in the vicinity of **H**.

Discharge:

Mayo River near Mayo	na
Mayo Lake near the Outlet	2.998 m
Wareham Lake at Headgate	14.3 m³/s



(iv)
Date: September 6, 1996
A28299 #243

NOTE:

- Continued vegetation development on upland and riparian surfaces.
- Generally stable channel conditions with few unvegetated gravel bars.

Discharge:

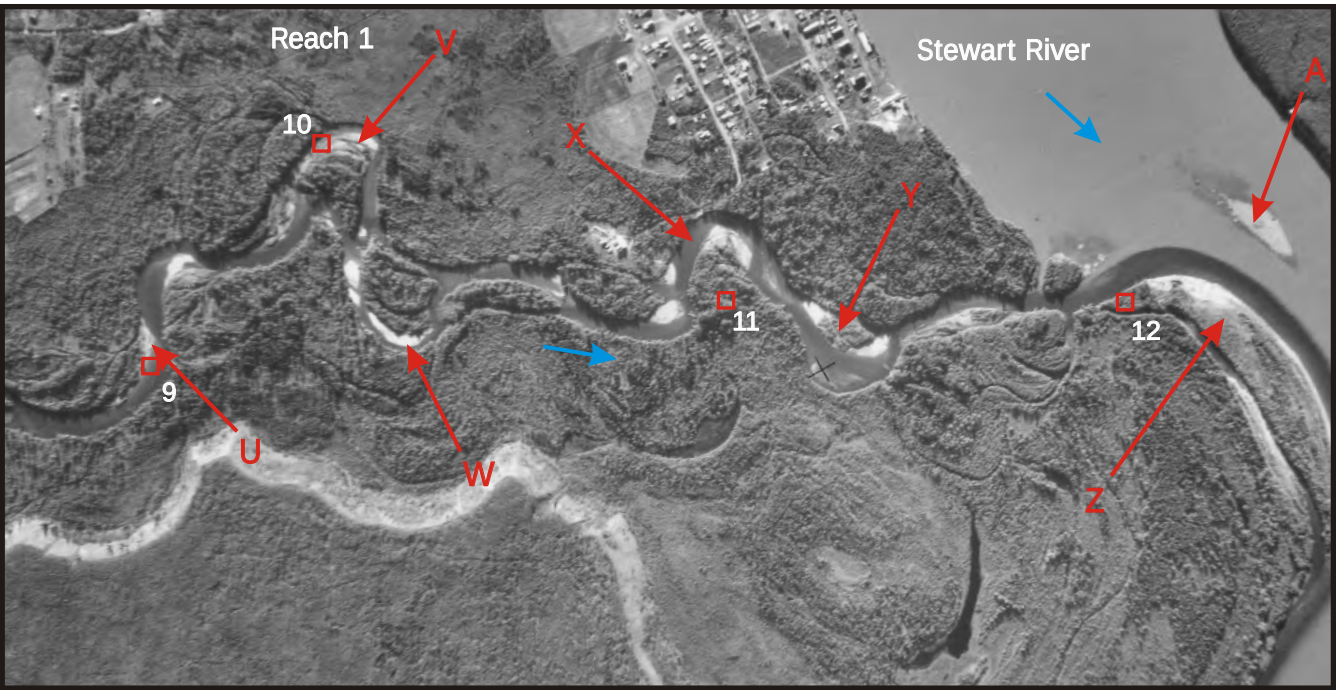
Mayo River near Mayo	na
Mayo Lake near the Outlet	3.002 m
Wareham Lake at Headgate	19.0 m³/s

■ Kilometer Mark

I Reach Break

Figure 4.3.6: Historical changes in channel morphology (reaches 2 and 3), Mayo River.

M. MILES AND



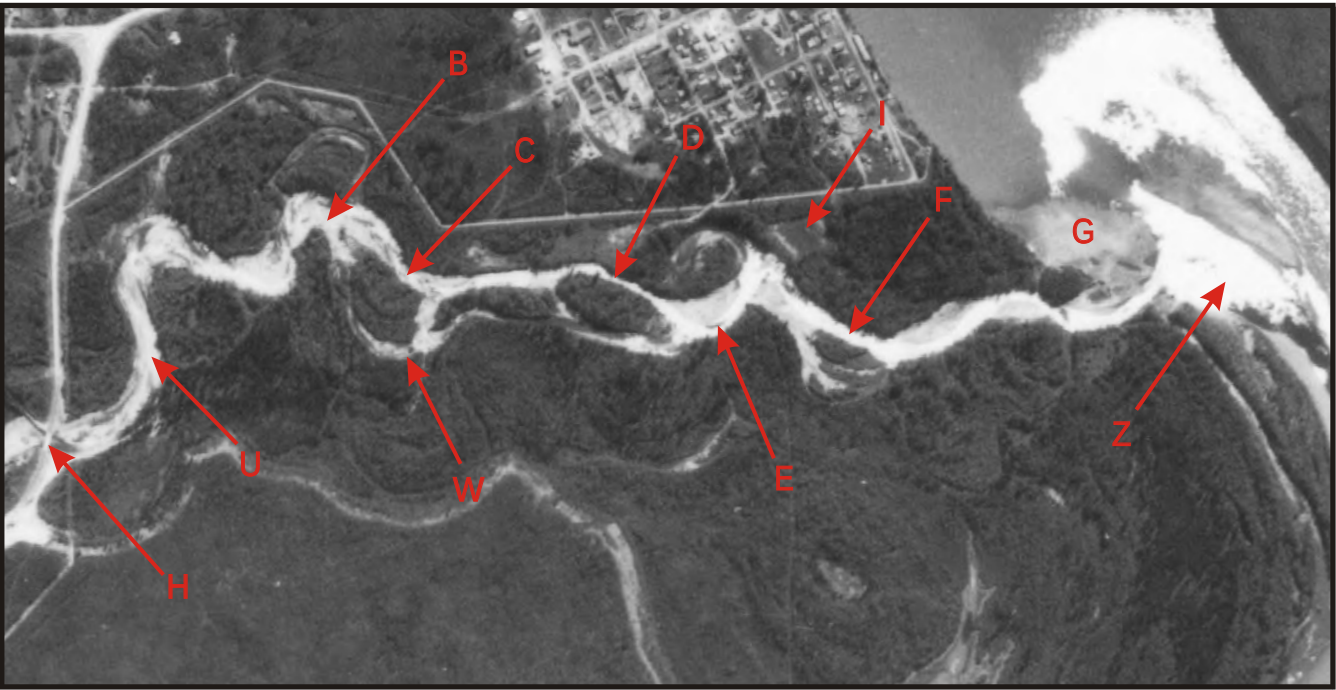
(i)
Date: July 27, 1949
A12133 #180

NOTE:

- Pre-regulation (1952) channel conditions.
- Local unvegetated instream sediment accumulation (eg. **U, V, W, X & Y**).
- Sediment deposits at confluence with Stewart River (**Z & A**).

Discharge:

Mayo River near Mayo	51.8 m³/s
Mayo Lake near the Outlet	na
Wareham Lake at Headgate	na
Stewart River at Mayo	951 m³/s
Stewart River near Mayo	na



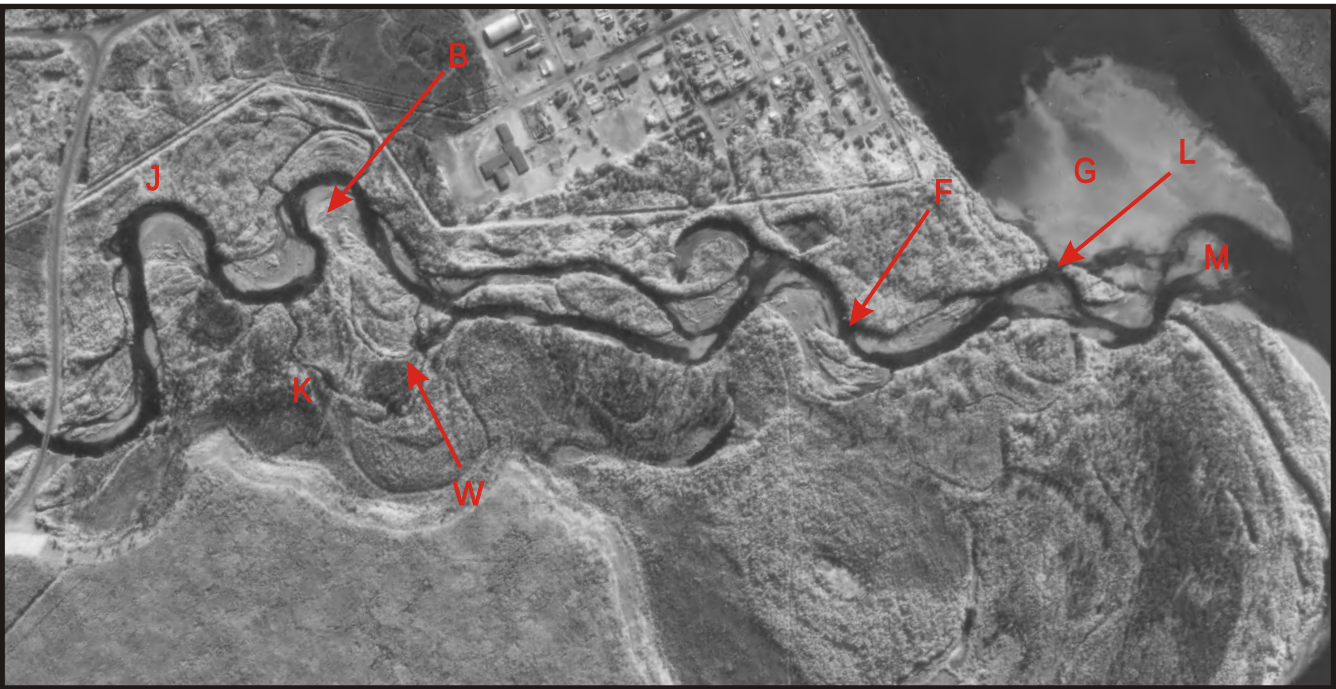
(ii)
Date: July 21, 1966
A19604 #26

NOTE:

- Bridge construction (**H**).
- Widespread post dam development of unvegetated instream sediment accumulations (eg. **U & W**).
- Numerous channel cutoffs (**B, C, D, E & F**).
- Enlarged sediment accumulations at the Stewart River confluence (**Z & G**).
- New pond at **I**.

Discharge:

Mayo River near Mayo	na
Mayo Lake near the Outlet	na
Wareham Lake at Headgate	na
Stewart River at Mayo	575 m³/s
Stewart River near Mayo	na



(iii)
Date: August 8, 1984
A26585 #40

NOTE:

- Widespread vegetation development on previously unvegetated instream gravel accumulations.
- Additional channel cutoffs or extensive channel shifting (eg. **J, K, B, W, F & L**).
- Continued presence of enlarged sediment deposits at Stewart River confluence (eg. **G & M**).

Discharge:

Mayo River near Mayo	na
Mayo Lake near the Outlet	2.998 m
Wareham Lake at Headgate	14.3 m³/s
Stewart River at Mayo	na
Stewart River near Mayo	4.217 m



(iv)
Date: September 6, 1996
A28299 #221

NOTE:

- New road **N** and probable pipeline crossing **O**. Sewage treatment plant **P**.
- Continued vegetation development along riparian areas.
- Extensive channel shifting at **J, K & Q**.
- New bar development in the vicinity of **L**.
- Breached pond at **I**.
- Continued presence of sediment deposits at the confluence with Stewart River **G**.

Discharge:

Mayo River near Mayo	na
Mayo Lake near the Outlet	3.002 m
Wareham Lake at Headgate	19.0 m³/s
Stewart River at Mayo	na
Stewart River near Mayo	5.145 m

■ Kilometer Mark ■ Reach Break

Figure 4.3.8: Historical changes in channel morphology (**Reach 1**), Mayo River.

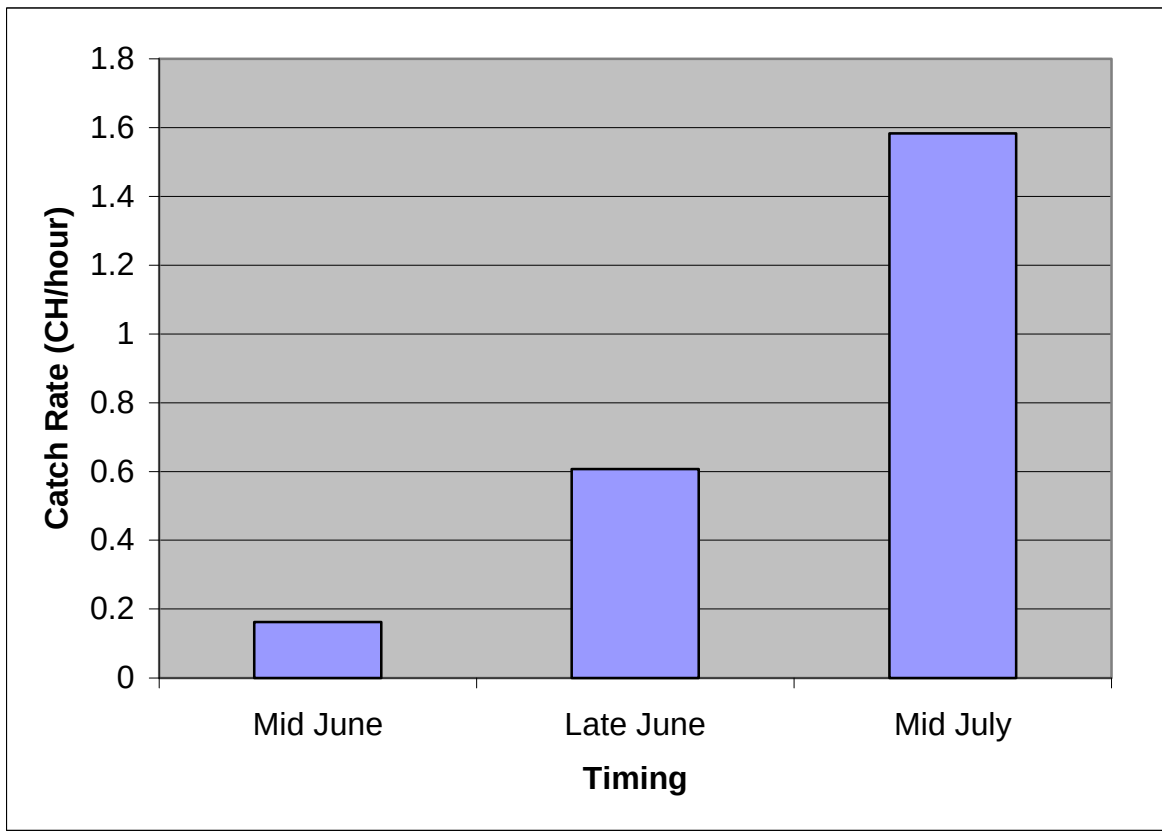
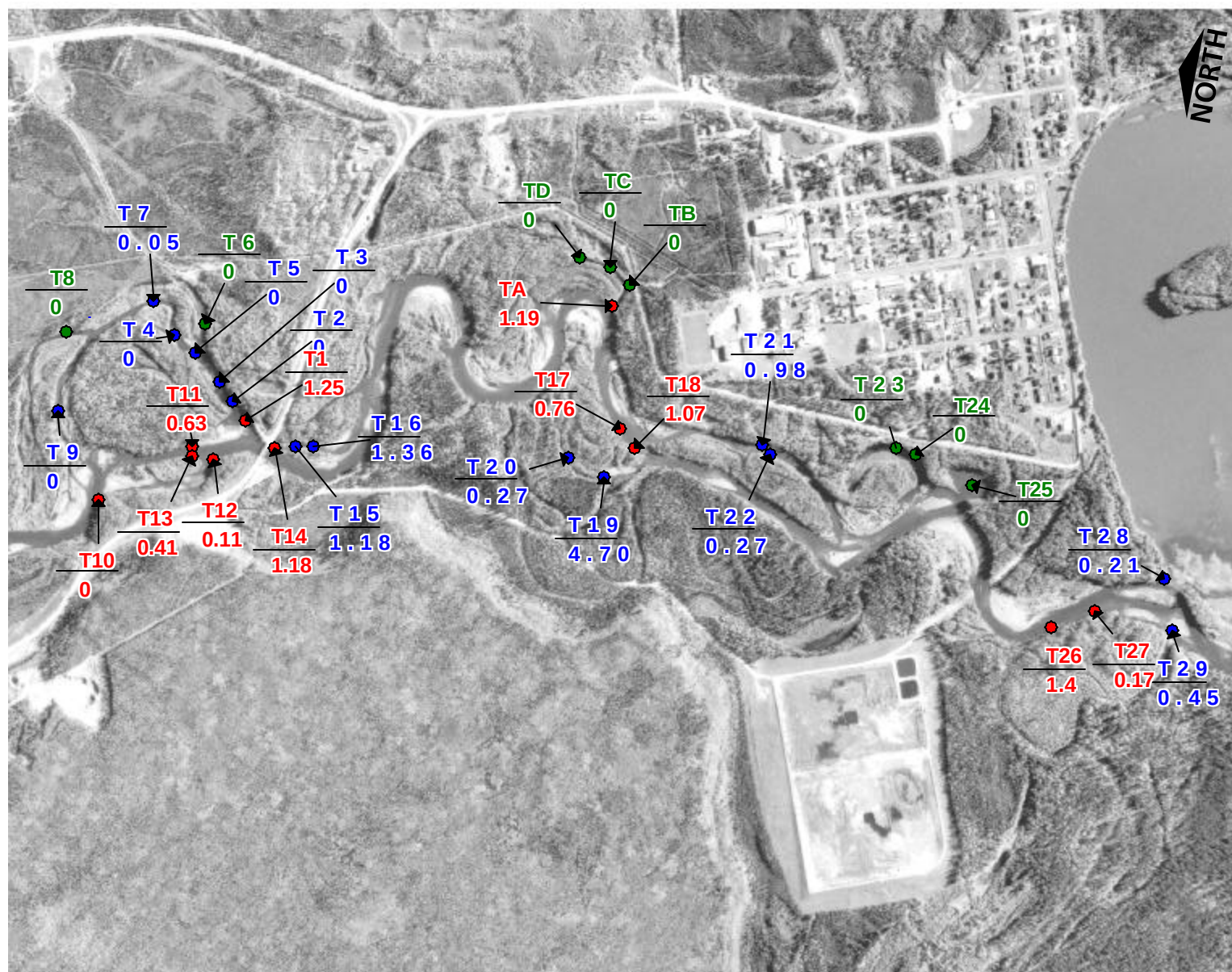


Figure 4.4.1. Average catch rate of juvenile chinook per hour per minnow trap in 1994 (Interpreted from Heron 1994).



Mayo River Juvenile Chinook Sampling July 2003

Date :December 4, 2003
Created by: Nathan Ferguson

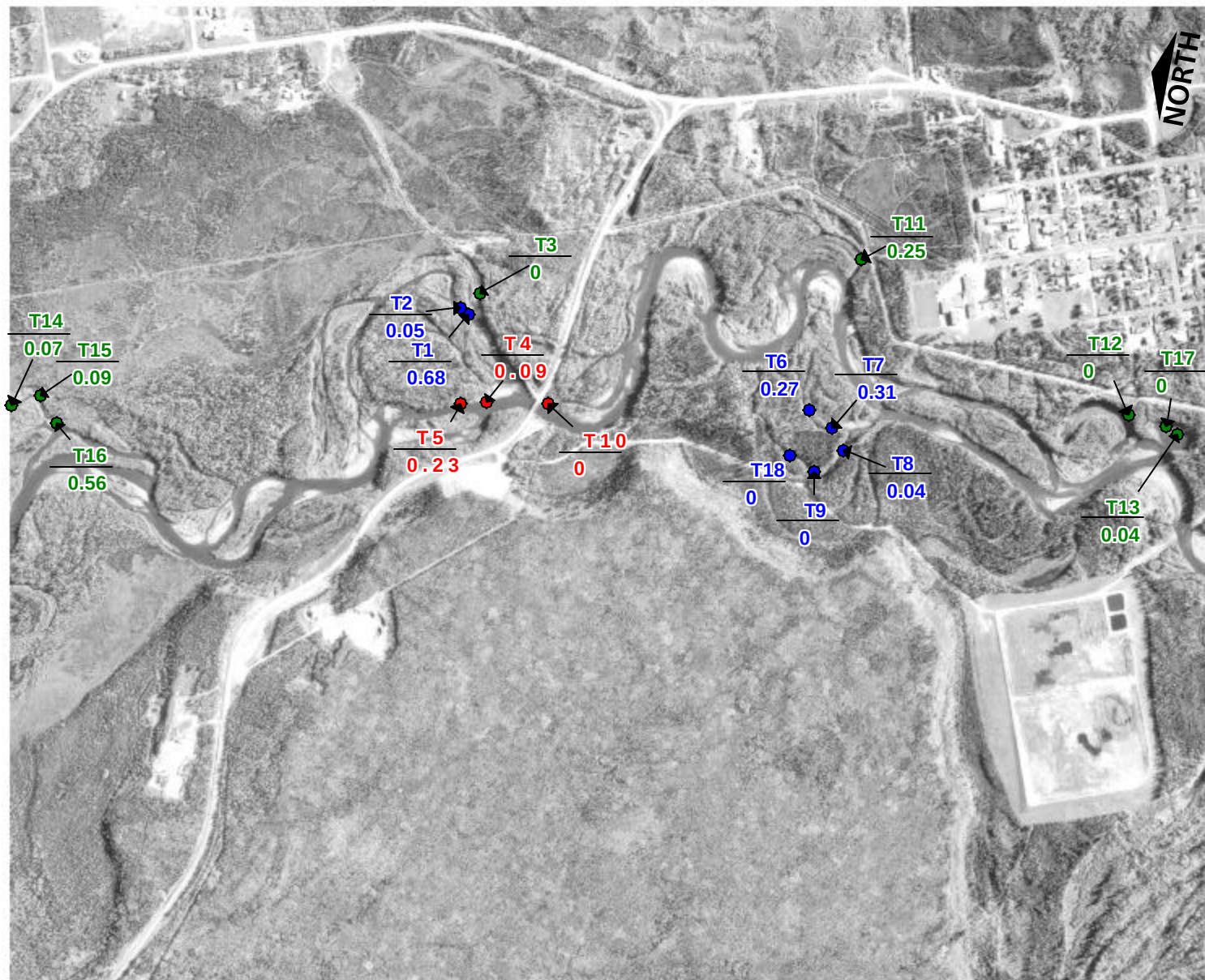
LEGEND

- Back Channel
- Side Channel
- Main Stem
- Trap #
○ Chinook/hour

Scale: 1:11,000
Meters



Figure 5.1.1. Minnow trap locations and results for the Mayo River for late July 2003.



Mayo River Juvenile Chinook Sampling September 30, 2003

Date :December 4, 2003
Created by: Nathan Ferguson

LEGEND

- Back Channel
- Side Channel
- Main Stem

Trap #
○ Chinook/hour

Scale: 1:11,000

Meters



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SUSTAINABLE RESOURCE MANAGEMENT

Figure 5.1.2. Minnow trap locations and results on the Mayo River for late September 2003.

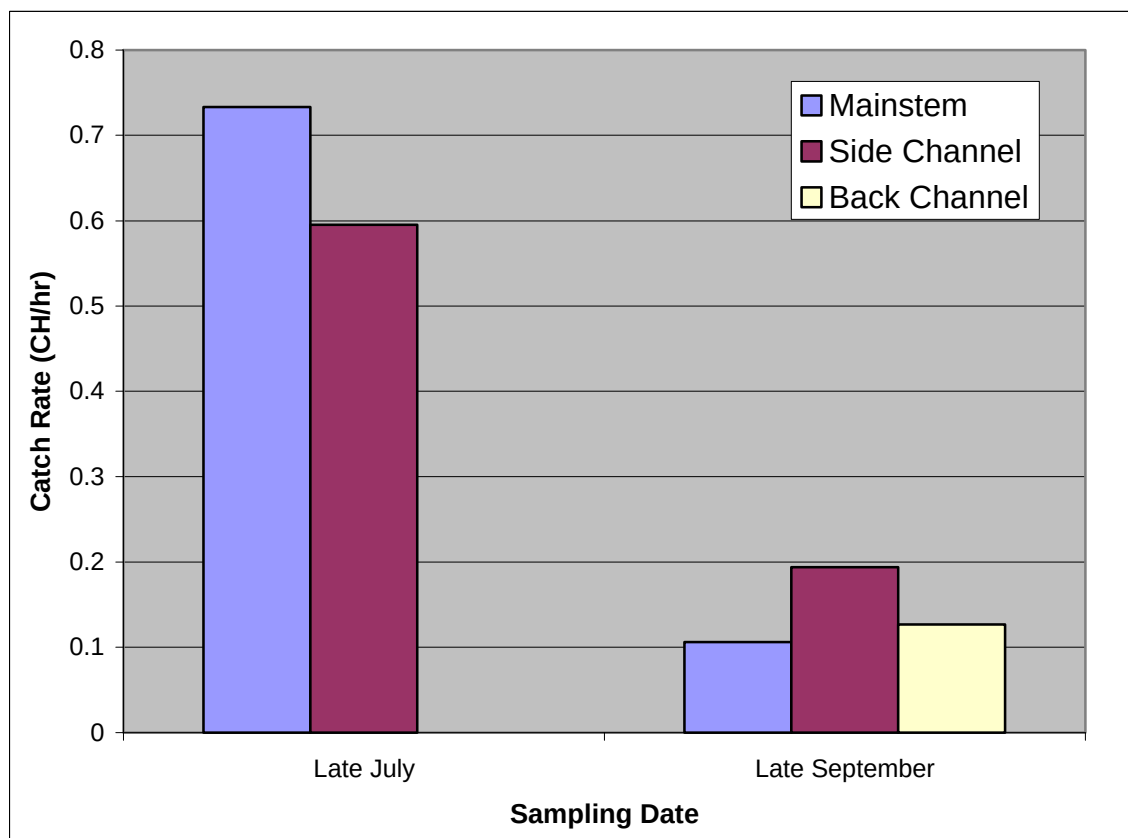


Figure 5.1.3 Catch rate within each habitat type.

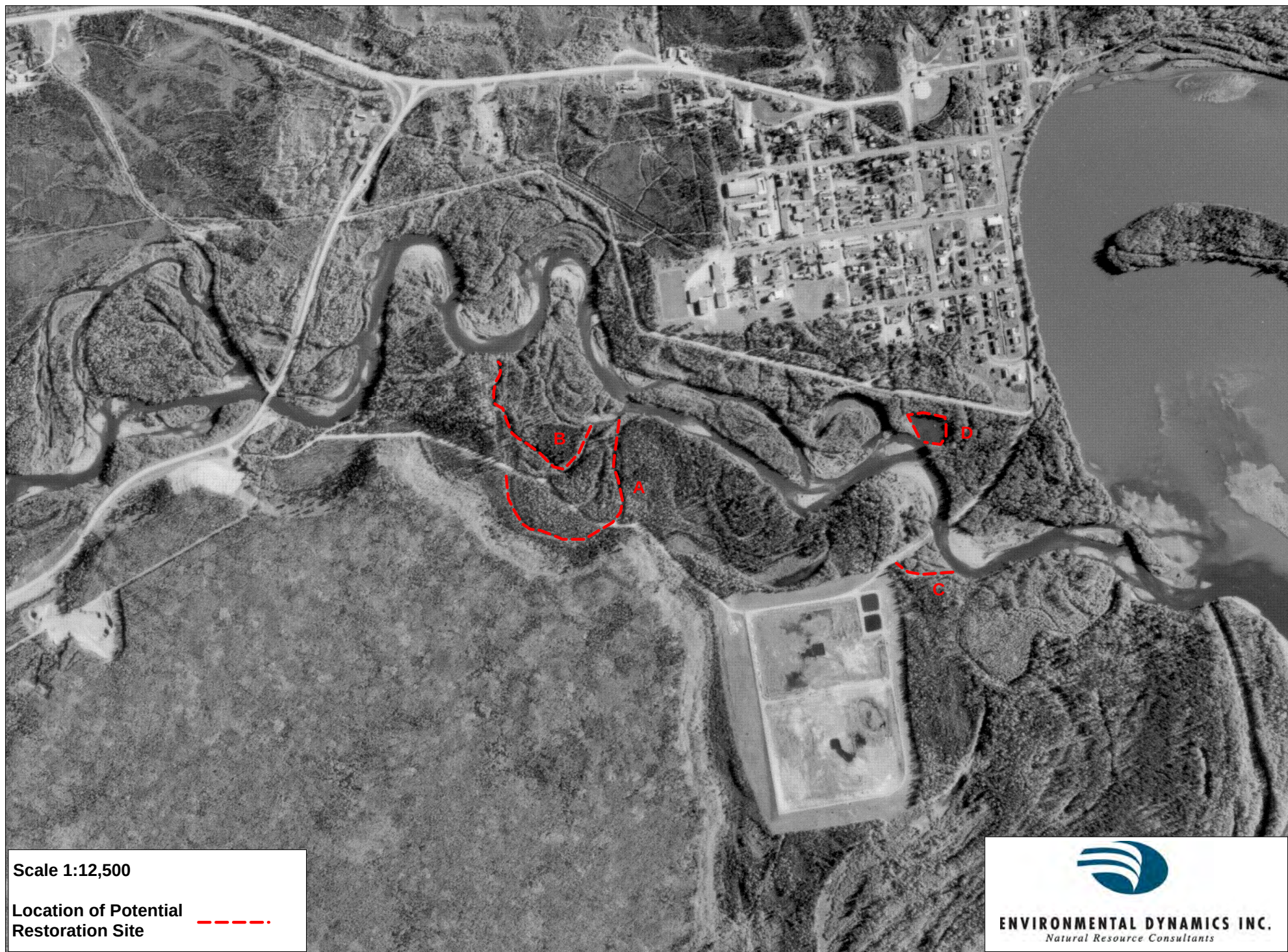


Figure 6.2.1 Location of potential restoration sites downstream of the highway crossing.



Figure 6.2.2 Location of potential restoration sites upstream of the highway crossing.

APPENDIX B: Minnow Trap Data

Lower Mayo River Channel Assessment-Minnow Trap Data - July 27-30, 2003

Trap #	Habitat	Time in	Time out	Total hours in	Size range (mm)	Temp C	Photo #	CH/HR	Other Species
A	main stem	1707	1630	23.5	50-70	18.9	69,72	1.191489	
B	back channel	1707	1630	23.5		18.5	70-71	0	
C	back channel	1707	1630	23.5		18.5	70-71	0	
D	back channel	1707	1630	23.5		18.3	70-71	0	
1	side channel	1330	930	20	50-60	18	50	1.25	
2	side channel	1340	930	20				0	
3	side channel	1345	930	20		16.1	51	0	
4	side channel	1400	945	20		15.3	53	0	BB, CCG
5	side channel	1400	945	20		16.8	52	0	BB
6	back channel	1418	945	20		17.5	54-55	0	
7	side channel	1435	955	20	50	17.5	56-57	0.05	
8	back channel	1450	955	20		13.9	58	0	
9	side channel	1510	1010	19		12.3	59-60	0	
10	main stem	1516	1010	19		19	61-62	0	
11	main stem	1535	1010	19	60-75	18.9	63-64	0.631579	
12	main stem	1540	1010	19	50-70	19	65-66	0.105263	
13	main stem	1545	1130	19.5	45-75	19	67-68	0.410256	
14	main stem	1100	912	22	55-70	17		1.181818	
15	main stem	1100	915	22	50-75	17	82	1.181818	
16	side channel	1100	922	22	55-75	17.5	83	1.363636	
17	main stem	1112	945	22.5	50-75 (1 @ 90)	17	84	0.755556	
18	main stem	1120	950	22.5	65-75	17	85	1.066667	
19	side channel	1129	950	23	60-80	15	86	4.695652	
20	side channel	1135	1007	22.5	65-70	15	87	0.266667	
21	side channel	1140	1053	22.5	60-80			0.977778	
22	side channel	1140	1100	22.5	60-70	17.5	88-89	0.266667	
23	side channel	1150	1100	23		14.5	90	0	
24	side channel	1200	1100	23		14.5	91	0	
25	back channel	1215	1100	23		15	92	0	
26	main stem	1630	1640	24	70-80	17.5	93	1.375	
27	main stem	1640	1630	24	50-70	17.5	94	0.166667	NP (95 mm)
28	side channel	1645	1615	24	60-70	17.5	95	0.208333	
29	side channel	1650	1620	24.5	60-70	18.5	96	0.44898	

Average 0.533146

Average CH/HR

Side Channel
0.595482036

Back Channel
0

Main stem
0.733283012

Lower Mayo River Channel Assessment-Minnow Trap Data-September 28-Oct 1, 2003

Trap #	Habitat	Time in	Time out	Total hours in	# CHj	Size range (mm)	Temp ©	Photo #	CH/HR	Other species
1	Side channel	14:30	12:30	22	15	65-80	6	18.19.20	0.681818	
2	Side channel	14:35	12:35	22	1	80	5	21.22	0.045455	
3	Back Channel	14:40	12:40	22	0		6	23.24	0	
4	Main stem	14:50	12:50	22	2	60-75	5	25.26	0.090909	
5	Main stem	15:00	13:00	22	5	70-85	5	27.28	0.227273	CCG
6	Side channel	15:15	13:15	22	6	65-80	5	29.30	0.272727	
7	Side channel	15:27	13:57	22.5	7	60-80	5	31.32	0.311111	
8	Side channel	15:37	14:07	22.5	1	75	5	33.34	0.044444	
9	Side channel	15:41	14:40	23	0		6	35.36	0	
9	Side channel	15:41	14:41	23	0				0	
10	Main stem	15:55	15:55	24	0		5.5	37.38	0	
11	Back Channel	16:00	16:00	24	6	65-80	5	39.4	0.25	
12	Back Channel	16:12	16:12	24	0				0	
17	Back Channel	16:16	16:46	24.5	0		6		0	
13	Back Channel	16:16	16:46	24.5	0		6		0	
13	Back Channel	16:16	16:46	24.5	1	75	6		0.040816	
14	Back Channel	17:30	12:30	43	3		5	41.42	0.069767	CCG
15	Back Channel	17:30	12:30	43	4		5	41.42	0.093023	
16	Back Channel	17:30	12:30	43	24		5	41.42	<u>0.55814</u>	

Average **0.141341**

Average CH/HR

Side

0.193651

Main stem

0.106061

Back Channel

0.126468